

Descriptive Statistics

2025-11-07

1. Data Overview

1.1 Description of the Data Gathering Process

The data considered in the project is from OpenAlex. We use a pair-based keyword search to identify articles which use or construct game-theoretical models. The keywords we are taken from two lists: 1. A list of JEL codes defining the subject of study, i.e. game theory. 2. A list of terms describing game setup which ensures that we only cover papers employing or constructing game-theoretic models. For each pair of keywords, we identify articles based on the appearance of keyword pairs in the title or abstract of the paper (maybe we also want to consider full texts? This would increase the number of articles drastically.) Currently, we focus only on articles and reviews written in English (Should we include books?). We drop articles from the corpus without any reference information. As the number of articles is rather low prior to 1980, we choose this year as the starting point for our analysis. In this version ("v4"), we furthermore consider only articles which have journal information and whose journals are listed by the USNSF in order to assign disciplines to journals similar to Truc et al. (2023).

1.2 Search Terms

Table 1: List of Query Terms

Term	Term
game theory	equilibrium
game-theoretic	equilibria
cooperative game	solution concept
cooperative games	solution concepts
noncooperative game	payoff matrix
noncooperative games	pay-off matrix
non-cooperative game	payoff matrices
non-cooperative games	pay-off matrices
stochastic game	strategy
stochastic games	strategies
dynamic game	strategic
dynamic games	preference
evolutionary game	preferences
evolutionary games	player
repeated game	players
repeated games	
bargaining theory	
matching theory	

1.3 Descriptive Summary of the Data

Table 2: Descriptive Summary

Description	
Articles	17743.00
Year (Begin)	1980.00
Year (End)	2024.00
Average Citations per Article	37.54
Authors	31851.00
Articles per Author	1.65
Authors per Article	2.96

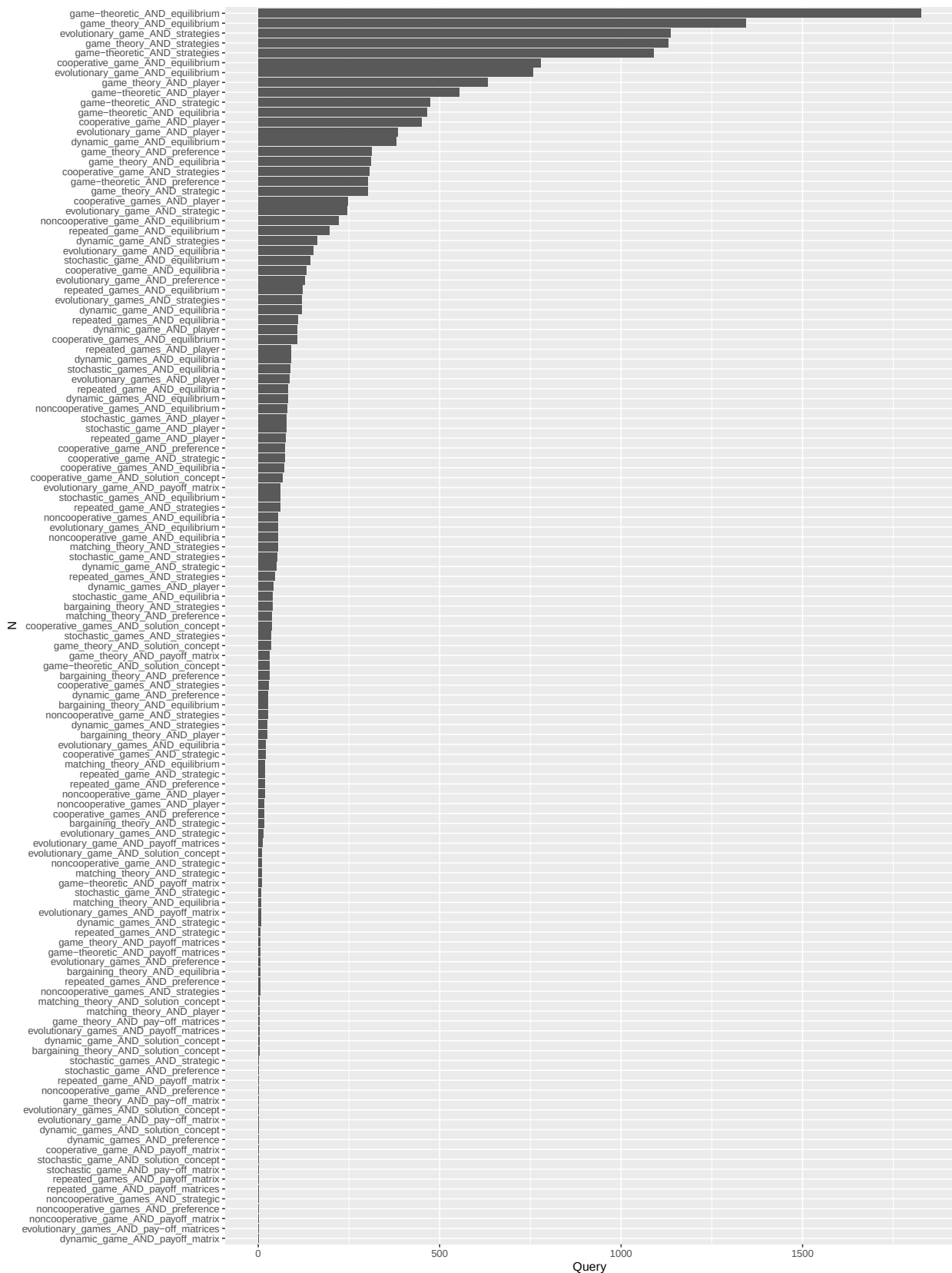
1.4 Top Articles

Table 3: Top Articles in Terms of Citations

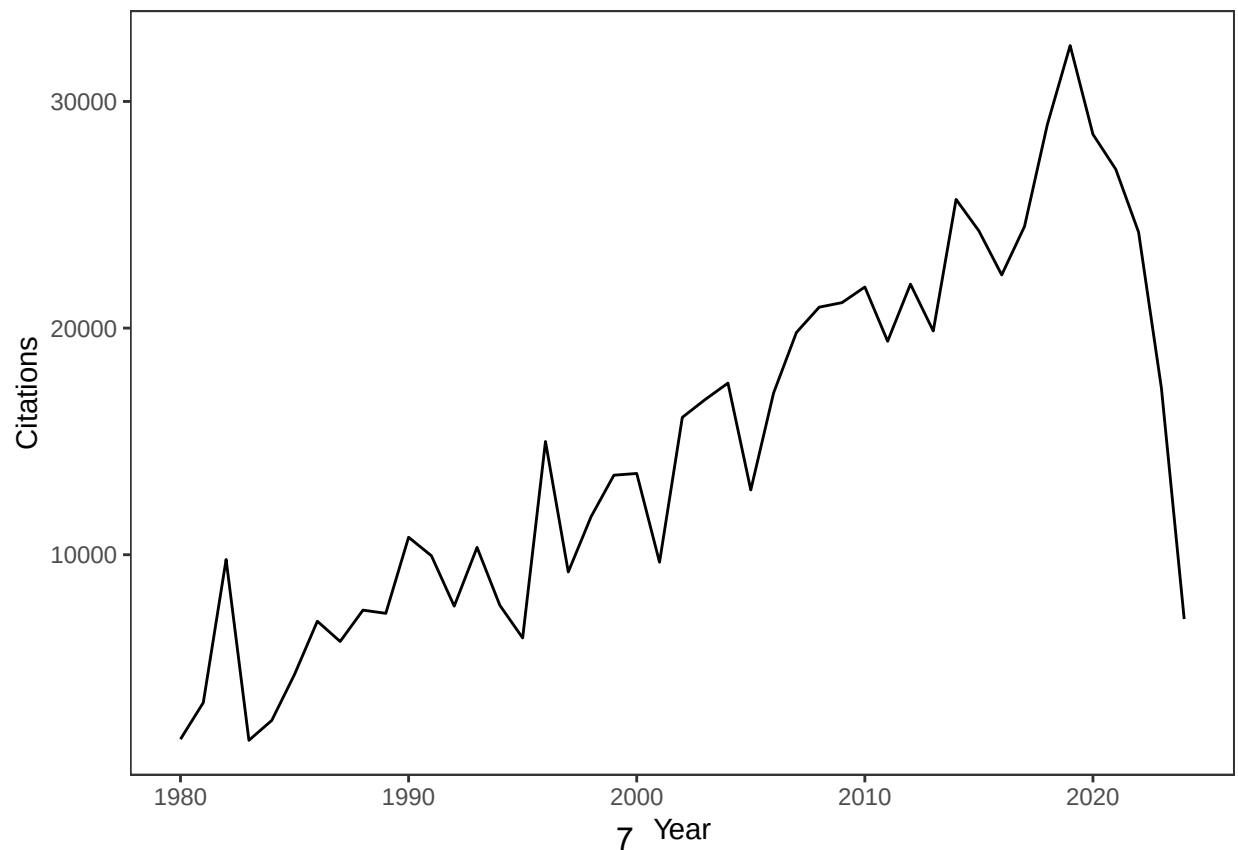
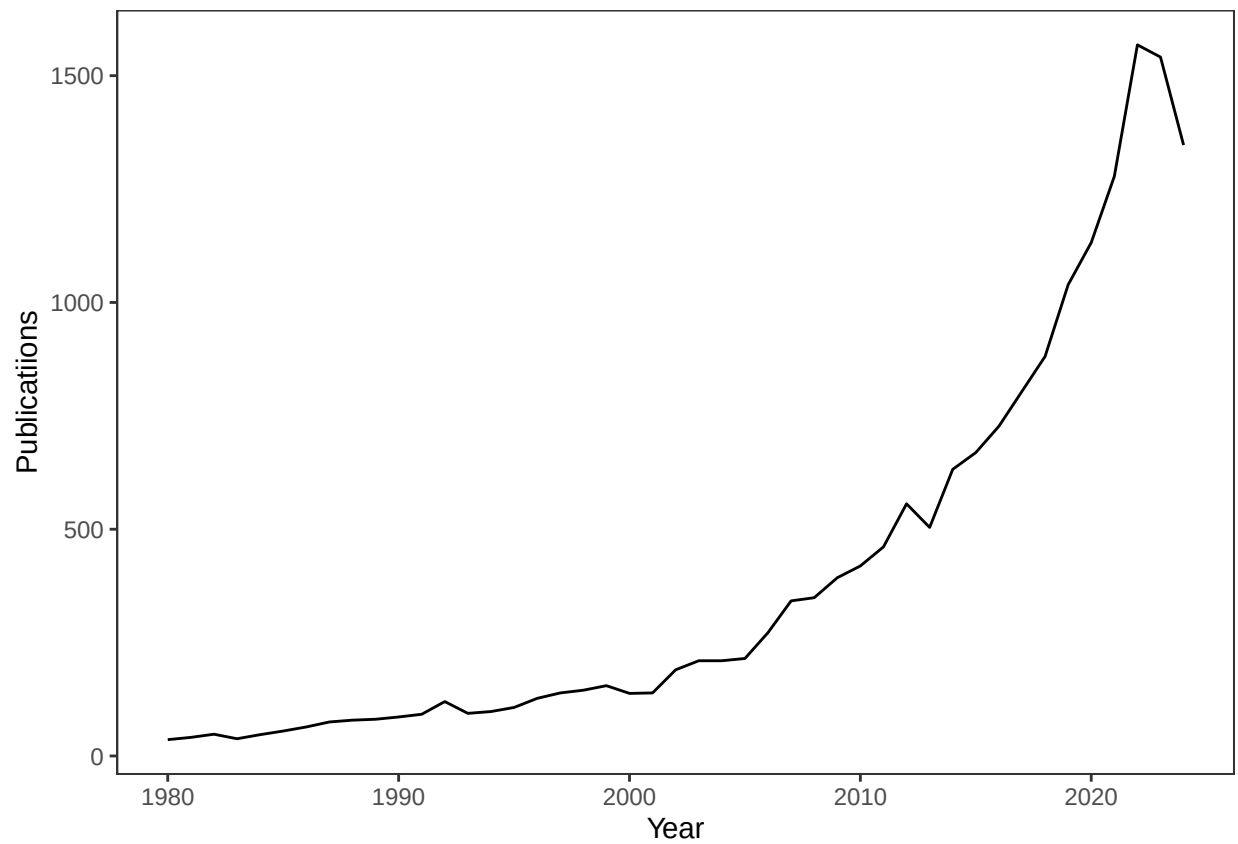
Authors	Title	Year	Citations	Field	Subfield	Discipline
David M. Kreps;Robert Wilson	Reputation and imperfect information	1982	3101	Decision Sciences	Management Science and Operations Research	Economics
Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	Autonomous Demand-Side Management Based on Game-Theoretic Energy Consumption Scheduling for the Future Smart Grid	2010	2711	Engineering	Electrical and Electronic Engineering	Engineering and Technology
Arvind Parkhe	STRATEGIC ALLIANCE STRUCTURING: A GAME THEORETIC AND TRANSACTION COST EXAMINATION OF INTERFIRM COOPERATION.	1993	2581	Business, Management and Accounting	Strategy and Management	Management
David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	Rational cooperation in the finitely repeated prisoners' dilemma	1982	2512	Social Sciences	Safety Research	Economics
Drew Fudenberg;Eric Maskin	The Folk Theorem in Repeated Games with Discounting or with Incomplete Information	1986	2371	Decision Sciences	Management Science and Operations Research	Economics
Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	A simple rule for the evolution of cooperation on graphs and social networks	2006	2155	Social Sciences	Sociology and Political Science	Biomedical Research
Charles F. Manski	Economic Analysis of Social Interactions	2000	2052	Economics, Econometrics and Finance	Economics and Econometrics	Economics
Martin Dufwenberg;Georg Kirchsteiger	A theory of sequential reciprocity	2003	1902	Social Sciences	Safety Research	Economics
Ido Erev;Alvin E. Roth	Predicting How People Play Games: Reinforcement Learning in Experimental Games with Unique, Mixed Strategy Equilibria	1998	1822	Social Sciences	Safety Research	Economics
Paul Milgrom;John Roberts	Rationalizability, Learning, and Equilibrium in Games with Strategic Complementarities	1990	1816	Decision Sciences	Management Science and Operations Research	Economics
Daniel Friedman	Evolutionary Games in Economics	1991	1797	Economics, Econometrics and Finance	Economics and Econometrics	Economics

Barry R. Weingast	The Political Foundations of Democracy and the Rule of the Law	1997	1739	Economics, Econometrics and Finance	Economics and Econometrics	Political Science
Mart R. Gross	Alternative reproductive strategies and tactics: diversity within sexes	1996	1681	Agricultural and Biological Sciences	Ecology, Evolution, Behavior and Systematics	Biology
Robert J. Aumann	Correlated Equilibrium as an Expression of Bayesian Rationality	1987	1603	Decision Sciences	Management Science and Operations Research	Economics
Amitav Mukherjee;S. Ali. A. Fakoorian;Jing Huang;A. Lee Swindlehurst	Principles of Physical Layer Security in Multiuser Wireless Networks: A Survey	2014	1427	Engineering	Electrical and Electronic Engineering	Engineering and Technology

1.5 Number of Articles per Query

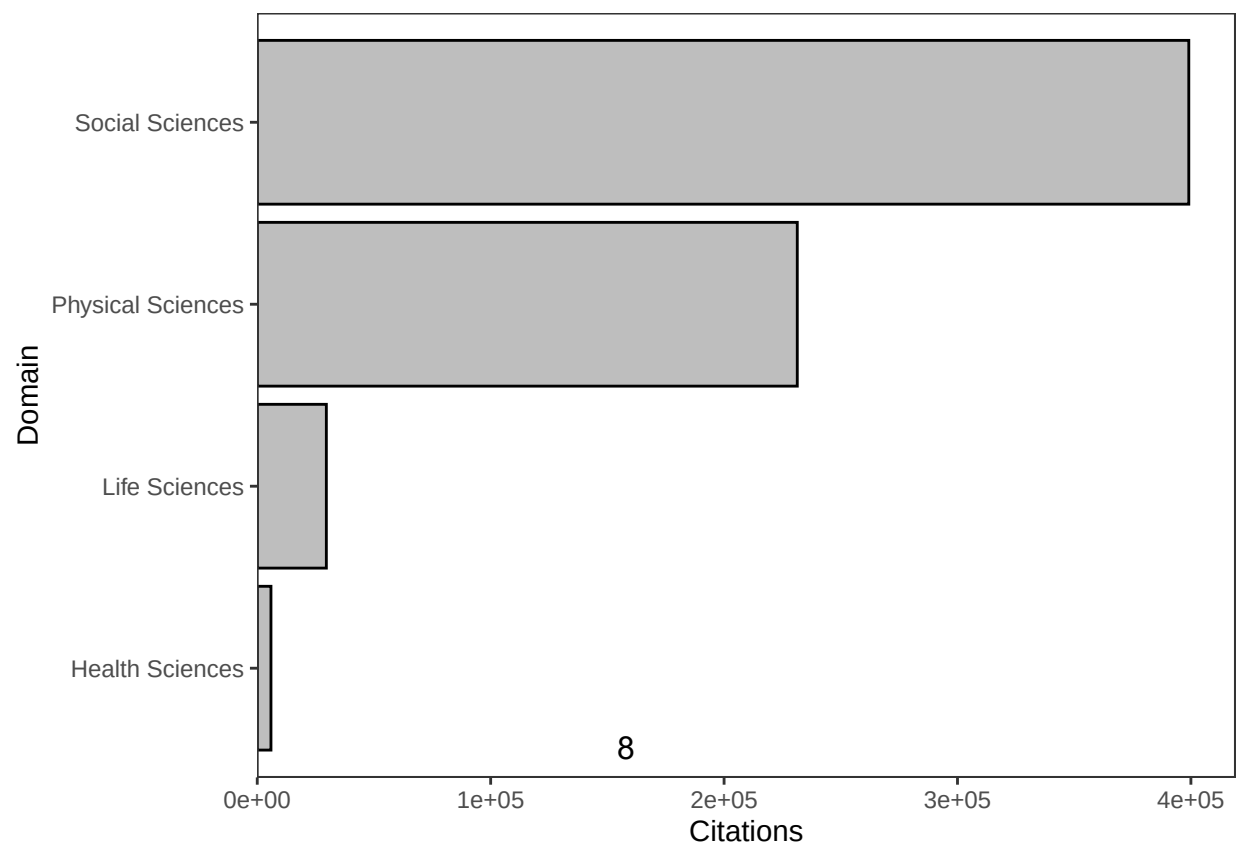
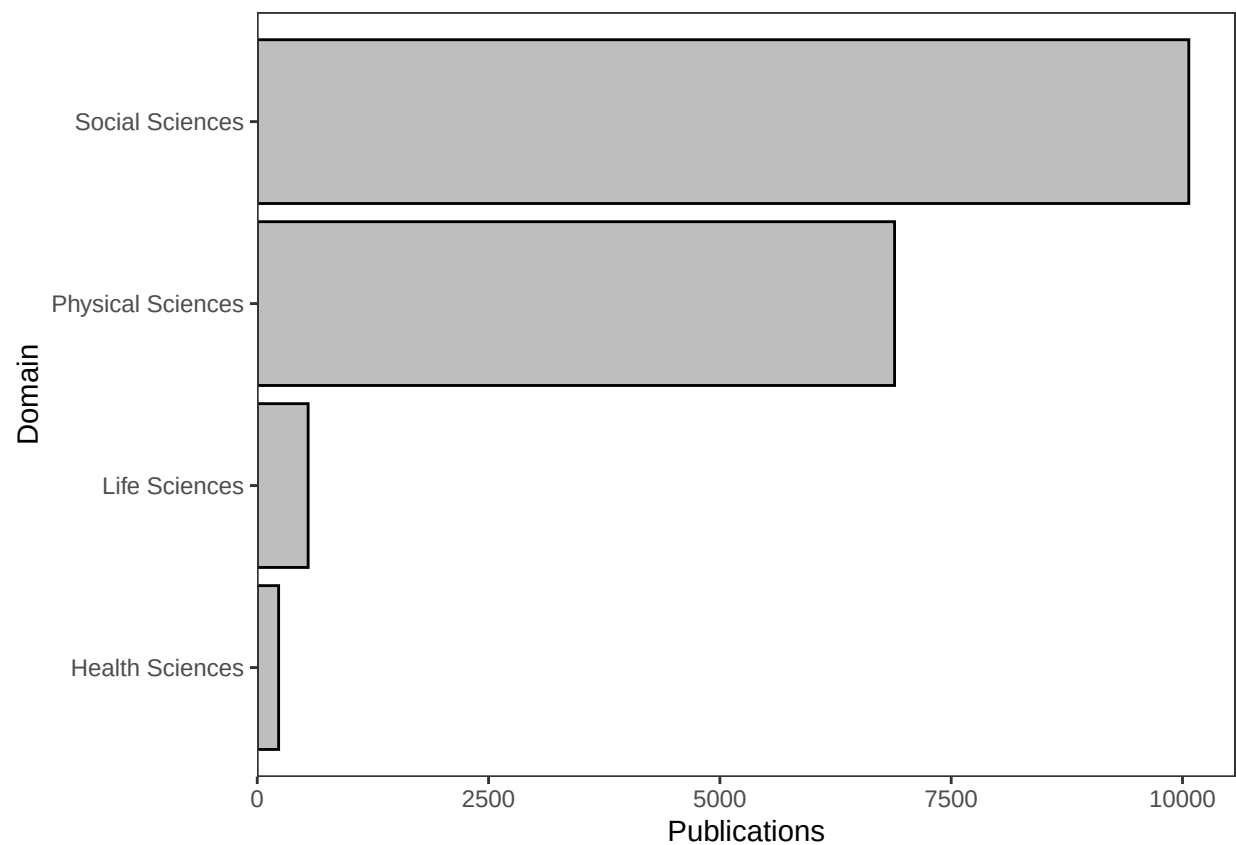


2. Publications over Time

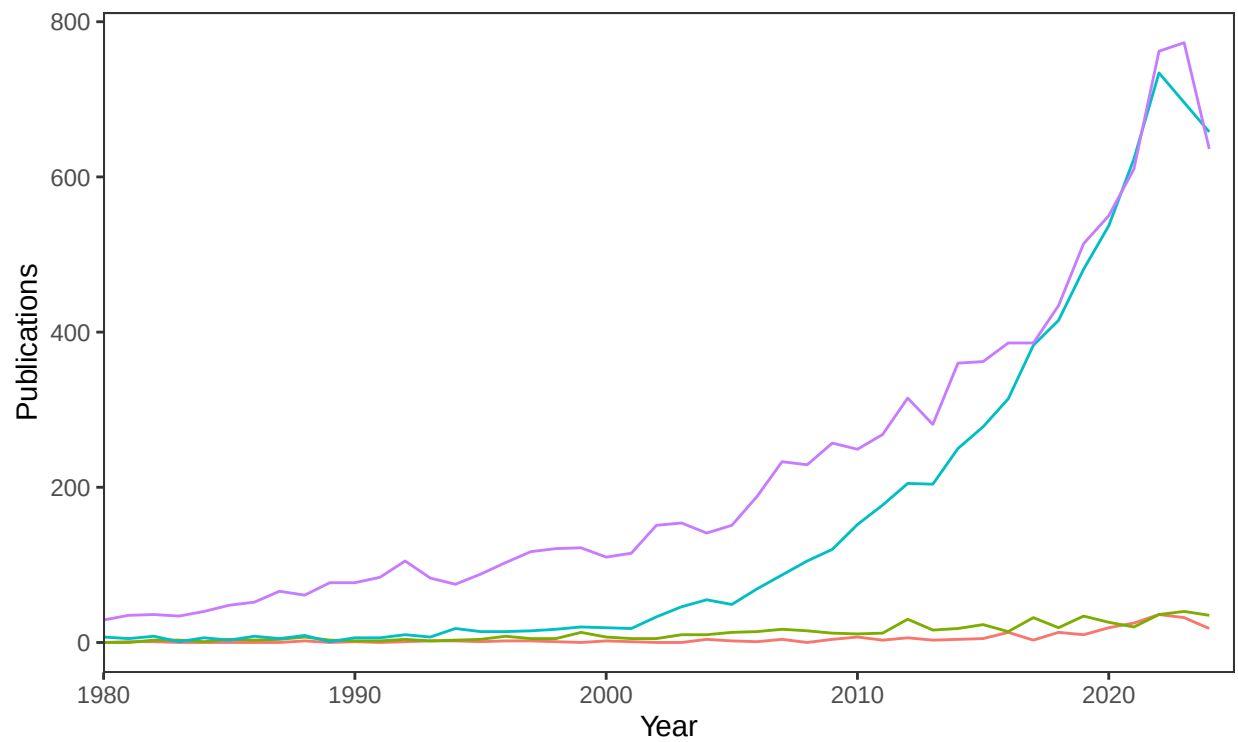


3. Domain Overview

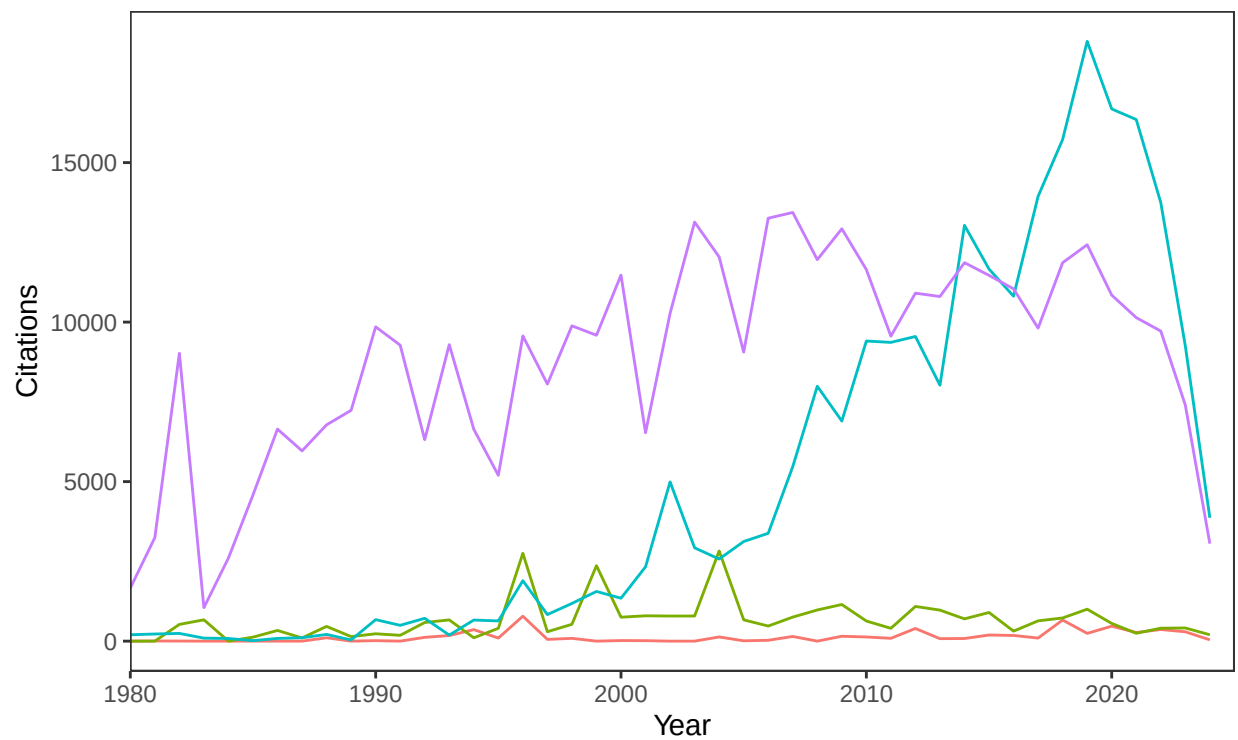
3.1 Top Domains



3.2 Domains over Time



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences

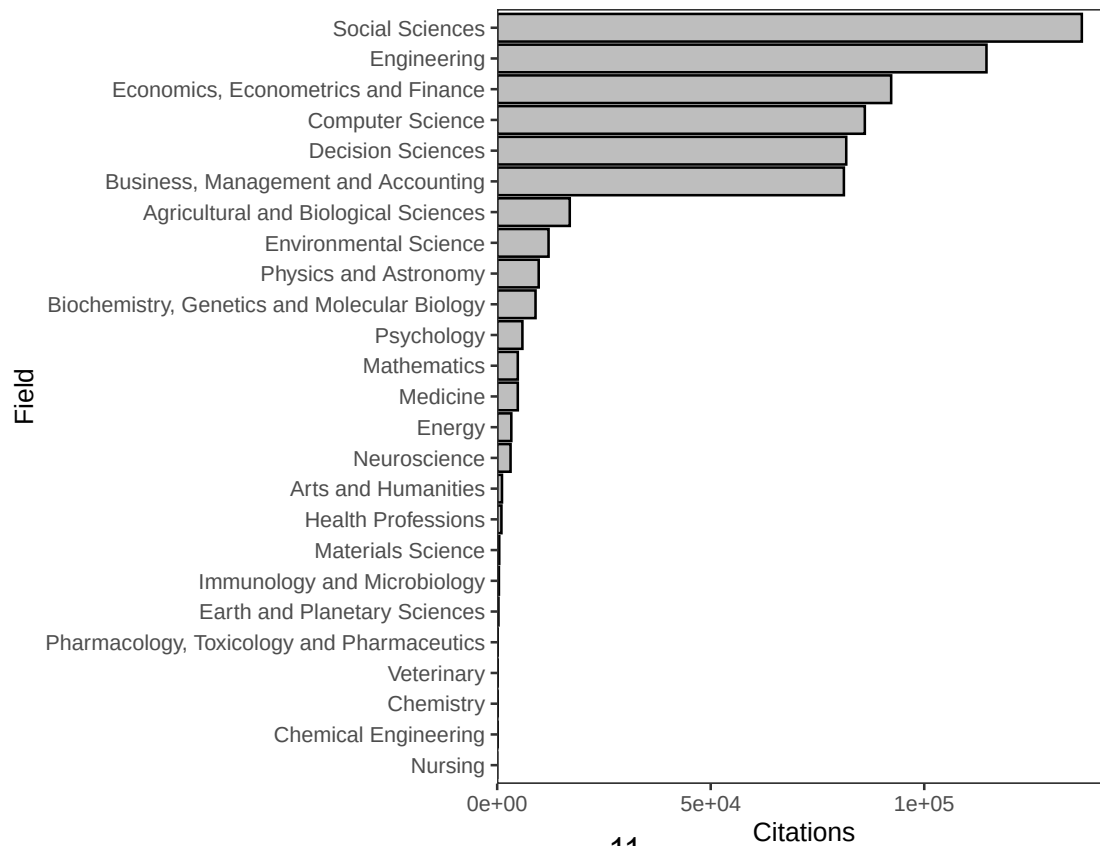
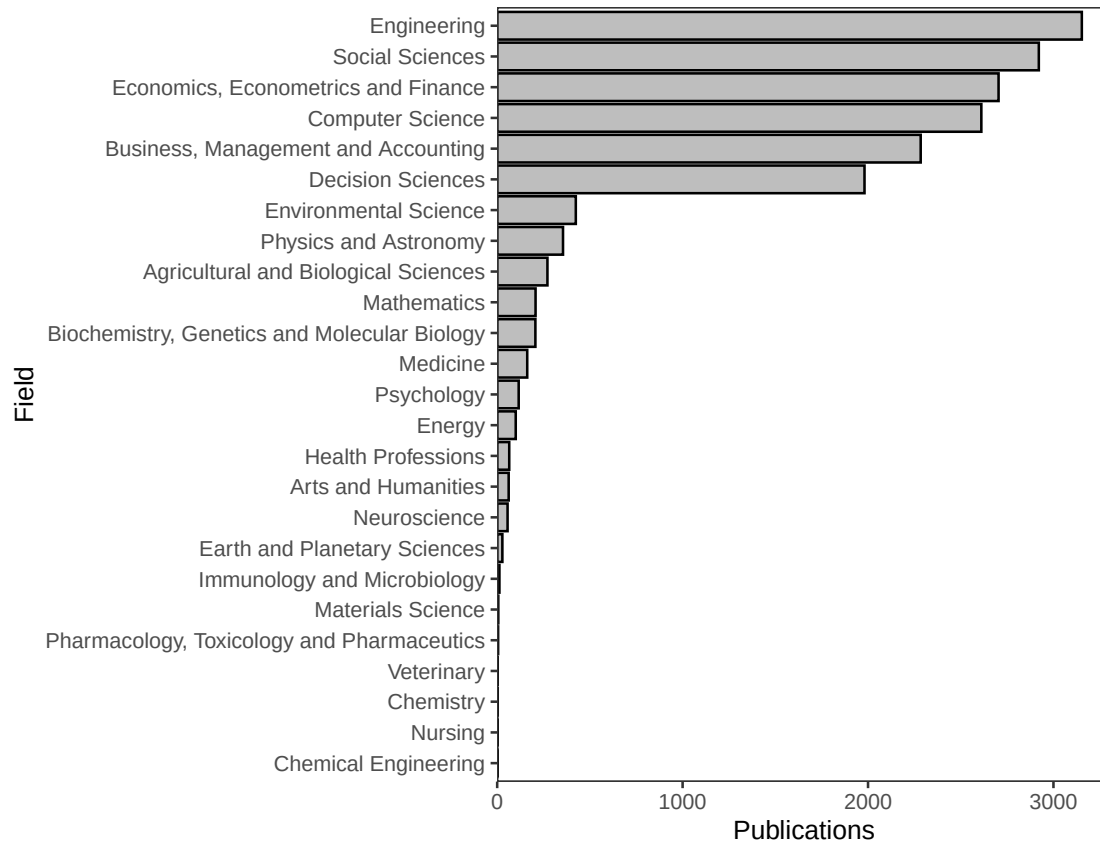
4. Field Overview

4.1 List of Fields

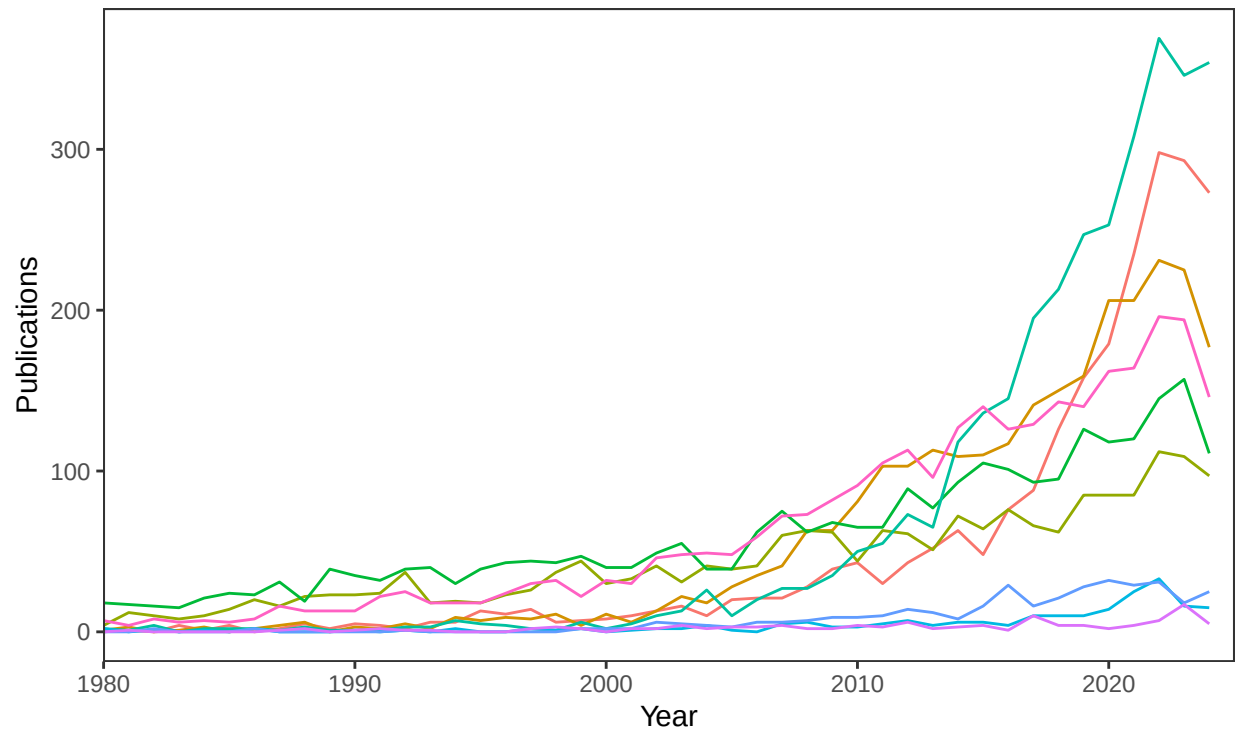
Table 4: List of Fields by OpenAlex.

Field
Agricultural and Biological Sciences
Arts and Humanities
Biochemistry, Genetics and Molecular Biology
Business, Management and Accounting
Chemical Engineering
Chemistry
Computer Science
Decision Sciences
Earth and Planetary Sciences
Economics, Econometrics and Finance
Energy
Engineering
Environmental Science
Health Professions
Immunology and Microbiology
Materials Science
Mathematics
Medicine
Neuroscience
Nursing
Pharmacology, Toxicology and Pharmaceutics
Physics and Astronomy
Psychology
Social Sciences
Veterinary

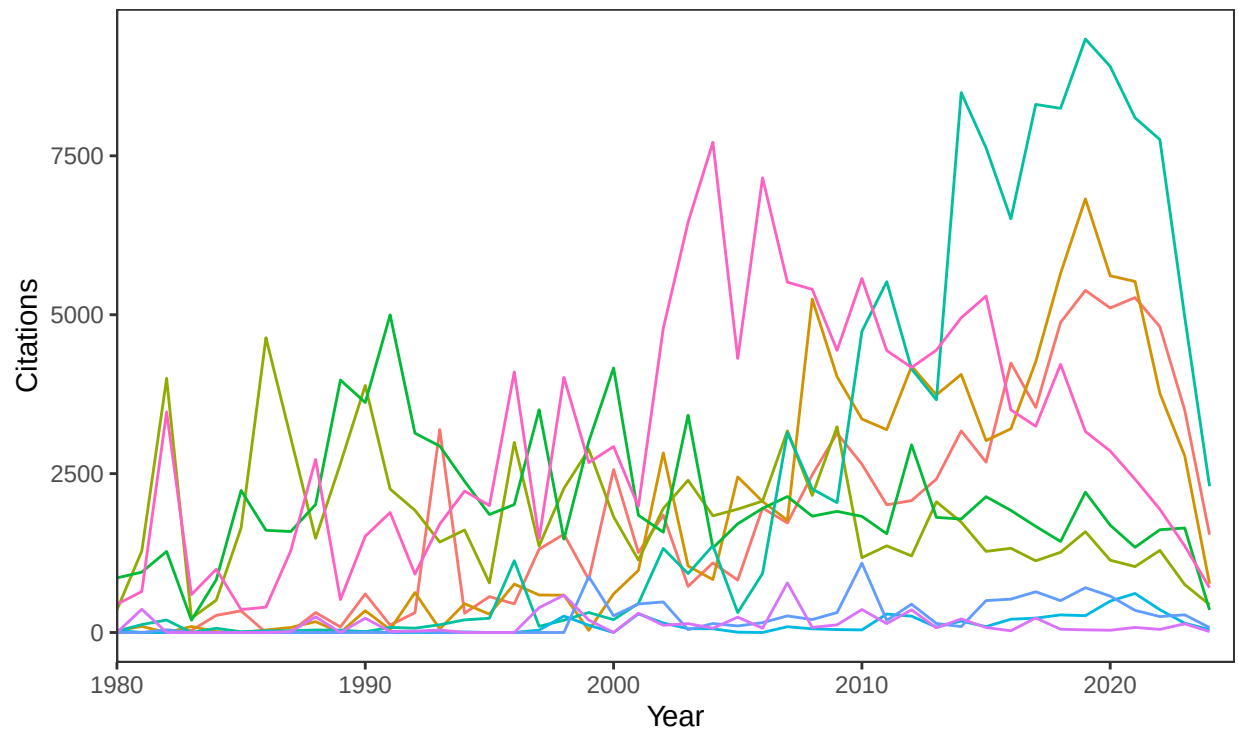
4.2 Top Fields



4.3 Fields over Time



Business, Management and Accounting Decision Sciences Engineering Physics and Astronomy Social Sciences
Computer Science Economics, Econometrics and Finance Mathematics Psychology



Business, Management and Accounting Decision Sciences Engineering Physics and Astronomy Social Sciences
Computer Science Economics, Econometrics and Finance Mathematics Psychology

4.4 Distribution of Fields across Queries

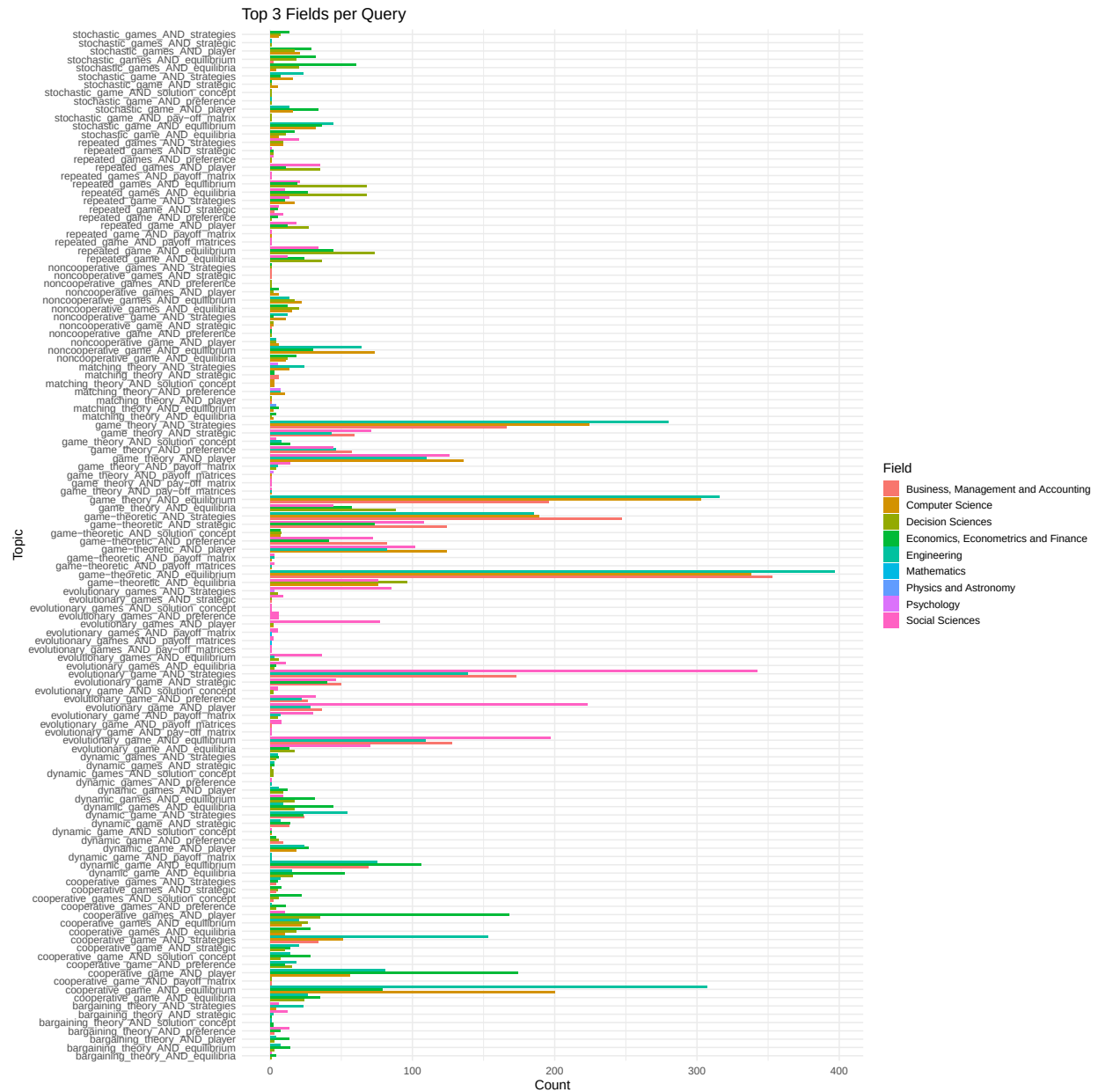
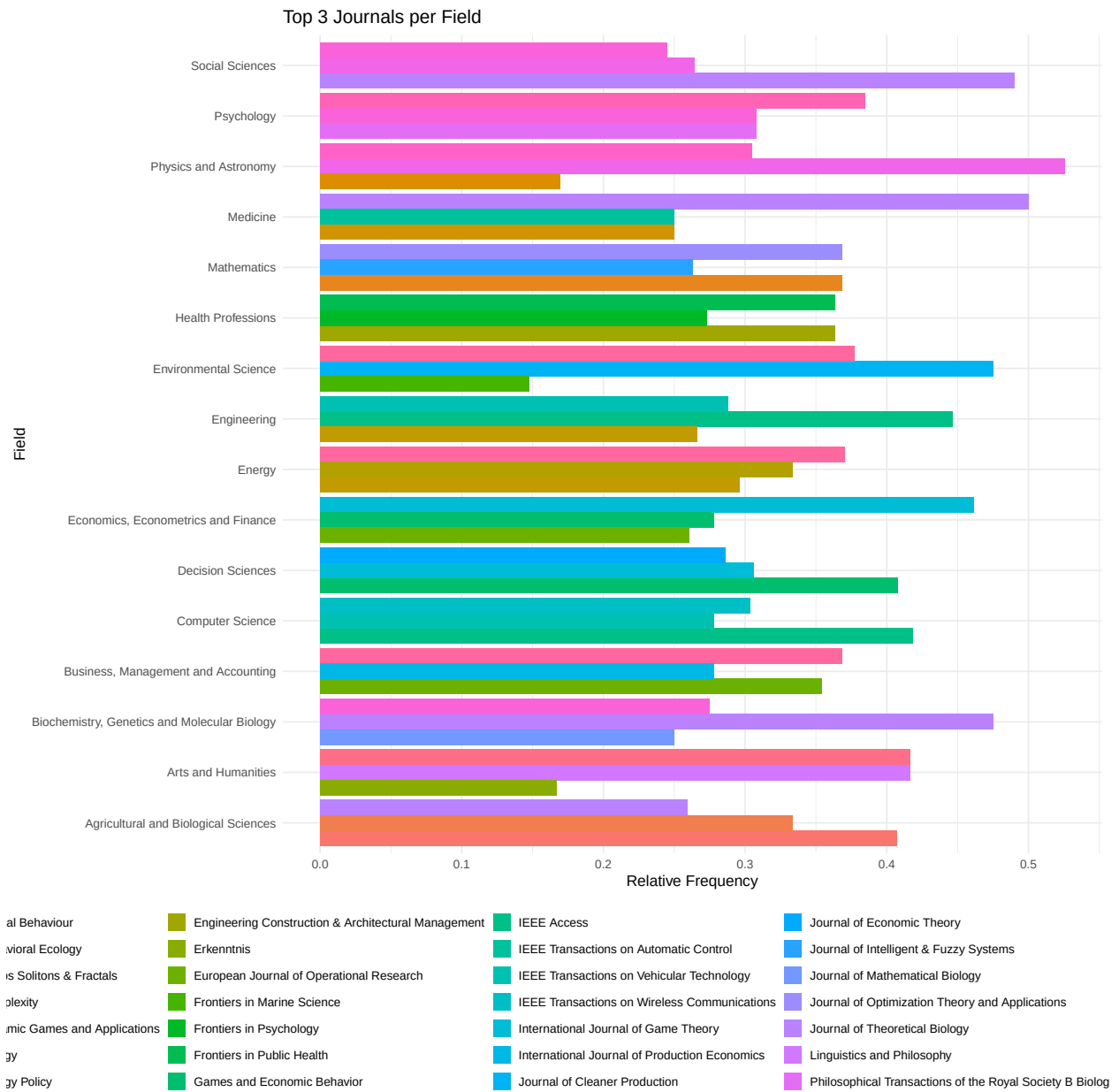


Table 5: Top Journals per Field

Field	Top Journals per Field (Descending Order)		
	Journal 1	Journal 2	Journal 3
Agricultural and Biological Sciences	Animal Behaviour (22)	Behavioral Ecology (18)	Journal of Theoretical Biology (14)
Arts and Humanities	Linguistics and Philosophy (5)	Synthese (5)	Erkenntnis (2)
Biochemistry, Genetics and Molecular Biology	Journal of Theoretical Biology (19)	PLoS ONE (11)	Journal of Mathematical Biology (10)
Business, Management and Accounting	Sustainability (126)	European Journal of Operational Research (121)	International Journal of Production Economics (95)
Computer Science	IEEE Access (113)	IEEE Transactions on Wireless Communications (82)	IEEE Transactions on Vehicular Technology (75)
Decision Sciences	Games and Economic Behavior (124)	International Journal of Game Theory (93)	Journal of Economic Theory (87)
Economics, Econometrics and Finance	International Journal of Game Theory (108)	Games and Economic Behavior (65)	European Journal of Operational Research (61)
Energy	Sustainability (10)	Energy Policy (9)	Energy (8)
Engineering	IEEE Access (141)	IEEE Transactions on Vehicular Technology (91)	Energy (84)
Environmental Science	Journal of Cleaner Production (29)	Sustainability (23)	Frontiers in Marine Science (9)
Health Professions	Engineering Construction & Architectural Management (4)	Frontiers in Public Health (4)	Frontiers in Psychology (3)
Mathematics	Chaos Solitons & Fractals (7)	Journal of Optimization Theory and Applications (7)	Journal of Intelligent & Fuzzy Systems (5)
Medicine	Journal of Theoretical Biology (10)	Dynamic Games and Applications (5)	IEEE Transactions on Automatic Control (5)
Physics and Astronomy	Physica A Statistical Mechanics and its Applications (31)	Quantum Information Processing (18)	Complexity (10)
Psychology	Scientific Reports (5)	PLoS ONE (4)	Philosophical Transactions of the Royal Society B Biological Sciences (4)
Social Sciences	Journal of Theoretical Biology (152)	Physica A Statistical Mechanics and its Applications (82)	PLoS ONE (76)

4.5 Distribution of Journals across fields



4.6 Top Articles per Field

Table 6: Top Articles per Field

Field	Top Articles per Field (Descending Order)		
	Authors	Title	No. Citations
Business, Management and Accounting	Arvind Parkhe	strategic alliance structuring: a game theoretic and transaction cost examination of interfirm cooperation.	2581
	Jan W. Rivkin	imitation of complex strategies	1292
	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	the interorganizational learning dilemma: collective knowledge development in strategic alliances	1077
Computer Science	Cem Saraydar;Narayan B. Mandayam;D.J. Goodman	efficient power control via pricing in wireless data networks	1332
	Wenbo Wang;Dinh Thai Hoang;Peizhao Hu;Zehui Xiong;Dusit Niyato;Ping Wang;Yonggang Wen;Dong In Kim	a survey on consensus mechanisms and mining strategy management in blockchain networks	940
	Xu Chen;Xu Chen	decentralized computation offloading game for mobile cloud computing	843
Decision Sciences	David M. Kreps;Robert Wilson	reputation and imperfect information	3101
	Drew Fudenberg;Eric Maskin	the folk theorem in repeated games with discounting or with incomplete information	2371
	Paul Milgrom;John Roberts	rationalizability, learning, and equilibrium in games with strategic complementarities	1816
Economics, Econometrics and Finance	Charles F. Manski	economic analysis of social interactions	2052
	Daniel Friedman	evolutionary games in economics	1797
	Barry R. Weingast	the political foundations of democracy and the rule of the law	1739

Engineering	Amir-Hamed Mohsenian-Rad; Vincent W. S. Wong; Juri Jatskevich; Robert Schober; Alberto Leon-Garcia	autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid	2711
	Amitav Mukherjee; S. Ali. A. Fakoorian; Jing Huang; A. Lee Swindlehurst	principles of physical layer security in multiuser wireless networks: a survey	1427
	Tamer Başar; P. Bernhard	$h/\sup \infty$ -Optimal control and related minimax design problems: a dynamic game approach	1066
		algal games: the vertical distribution of phytoplankton in poorly mixed water columns	294
Mathematics	Christopher A. Klausmeier; Elena Litchman	the determinacy of blackwell games	259
	Donald A. Martin	formulating an n-person noncooperative game as a tensor complementarity problem	174
	Zheng-Hai Huang; Liqun Qi	quantum strategies	744
Physics and Astronomy	Ramasuri Narayanam; Y. Narahari	a shapley value-based approach to discover influential nodes in social networks	373
	Simon C. Benjamin; Patrick Hayden	multiplayer quantum games	309
Psychology	Herbert Gintis	a framework for the unification of the behavioral sciences	411
	Geoffrey F. Miller; Peter M. Todd	mate choice turns cognitive	368
	Calvin B. Harley	learning the evolutionarily stable strategy	366
Social Sciences	David M. Kreps; Paul Milgrom; John Roberts; Robert Wilson	rational cooperation in the finitely repeated prisoners' dilemma	2512
	Hisashi Ohtsuki; Christoph Hauert; E. James Lieberman; Martin A. Nowak	a simple rule for the evolution of cooperation on graphs and social networks	2155
	Martin Dufwenberg; Georg Kirchsteiger	a theory of sequential reciprocity	1902

5. Subfield Overview

5.1 List of Subfields

Table 7: List of Subfields by OpenAlex.

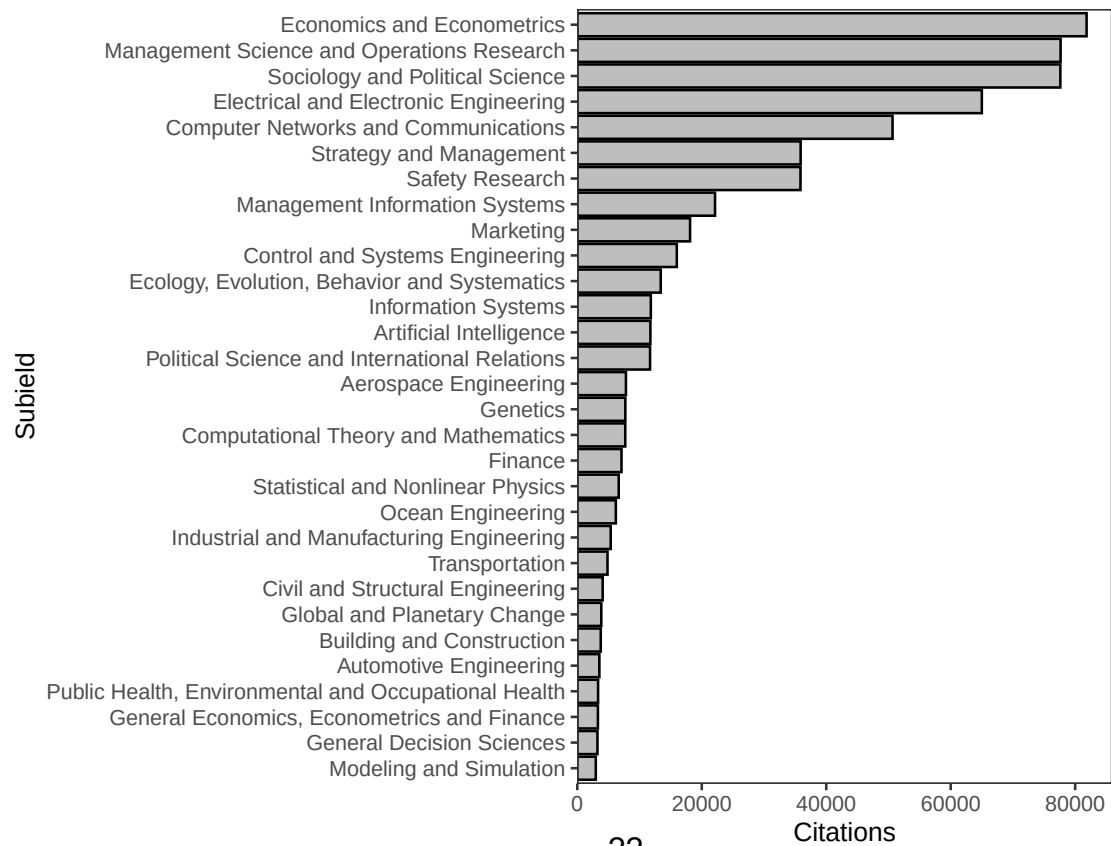
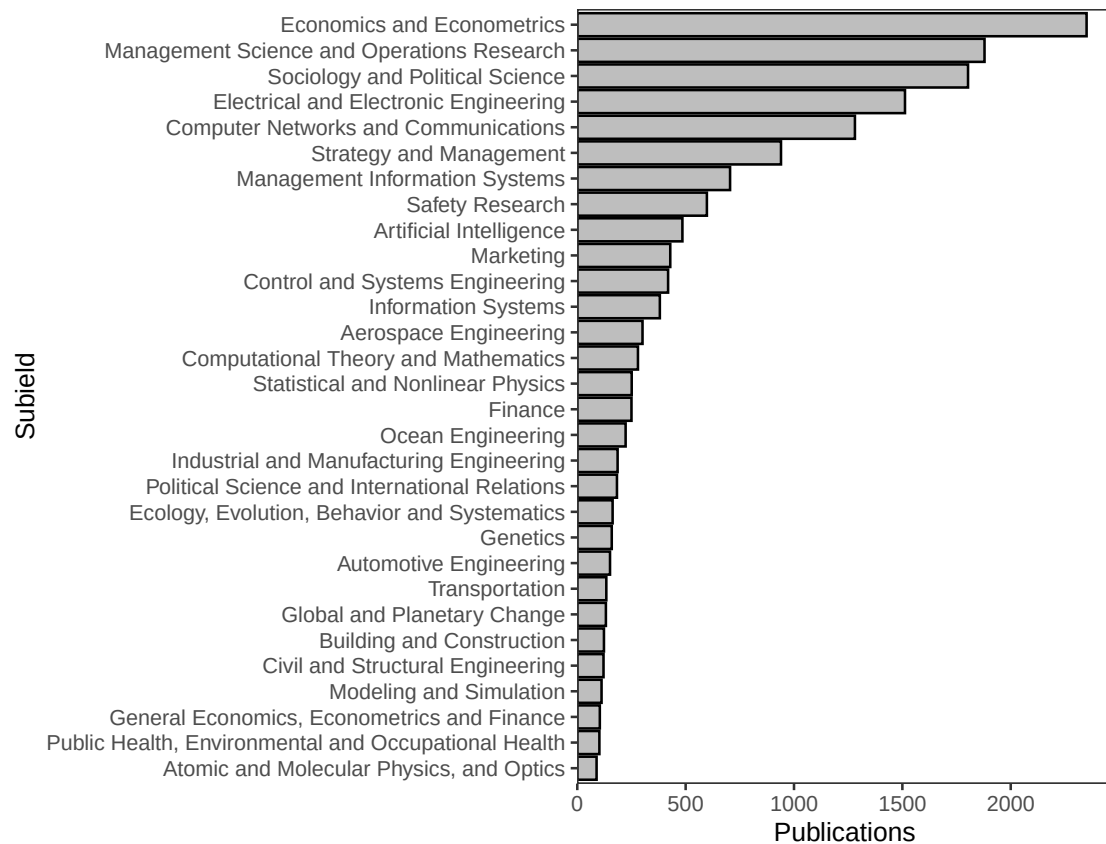
Subfield	No. Publications	No. Citations
Accounting	75	1753
Aerospace Engineering	301	7817
Aging	3	48
Agronomy and Crop Science	7	229
Algebra and Number Theory	2	14
Analytical Chemistry	1	10
Anthropology	3	35
Applied Mathematics	7	140
Applied Microbiology and Biotechnology	2	45
Applied Psychology	5	52
Aquatic Science	1	25
Artificial Intelligence	485	11747
Astronomy and Astrophysics	11	94
Atmospheric Science	4	17
Atomic and Molecular Physics, and Optics	89	2927
Automotive Engineering	151	3546
Biomedical Engineering	35	921
Building and Construction	123	3771
Business and International Management	21	301
Cancer Research	4	171
Cardiology and Cardiovascular Medicine	2	35
Cell Biology	2	57
Cellular and Molecular Neuroscience	2	166
Civil and Structural Engineering	121	4081
Clinical Psychology	13	488
Cognitive Neuroscience	51	2916
Communication	20	369
Complementary and alternative medicine	1	1
Computational Mathematics	8	420
Computational Mechanics	9	184
Computational Theory and Mathematics	280	7707
Computer Graphics and Computer-Aided Design	1	2
Computer Networks and Communications	1281	50657
Computer Science Applications	58	1562
Computer Vision and Pattern Recognition	84	1621
Control and Systems Engineering	419	15993

Cultural Studies	12	592
Demography	43	1544
Development	28	940
Developmental and Educational Psychology	24	1274
Discrete Mathematics and Combinatorics	3	19
Earth-Surface Processes	20	104
Ecological Modeling	3	54
Ecology	42	1383
Ecology, Evolution, Behavior and Systematics	163	13407
Economics and Econometrics	2350	81834
Education	28	487
Electrical and Electronic Engineering	1512	65001
Electronic, Optical and Magnetic Materials	2	34
Emergency Medical Services	8	158
Emergency Medicine	1	50
Endocrinology	2	98
Energy Engineering and Power Technology	5	130
Environmental Chemistry	3	14
Environmental Engineering	44	1424
Epidemiology	11	434
Experimental and Cognitive Psychology	29	1544
Family Practice	2	14
Finance	250	7103
Fluid Flow and Transfer Processes	1	16
Food Science	29	259
Gender Studies	24	1662
General Agricultural and Biological Sciences	15	174
General Decision Sciences	66	3238
General Economics, Econometrics and Finance	104	3318
General Energy	21	526
General Health Professions	17	107
General Psychology	1	224
Genetics	159	7721
Geography, Planning and Development	2	4
Geometry and Topology	16	362
Global and Planetary Change	132	3856
Hardware and Architecture	10	198
Health	5	36
Health Informatics	1	41
Health Information Management	3	15
Health, Toxicology and Mutagenesis	8	88
Hepatology	1	9
History	2	111
History and Philosophy of Science	8	339
Human-Computer Interaction	4	303

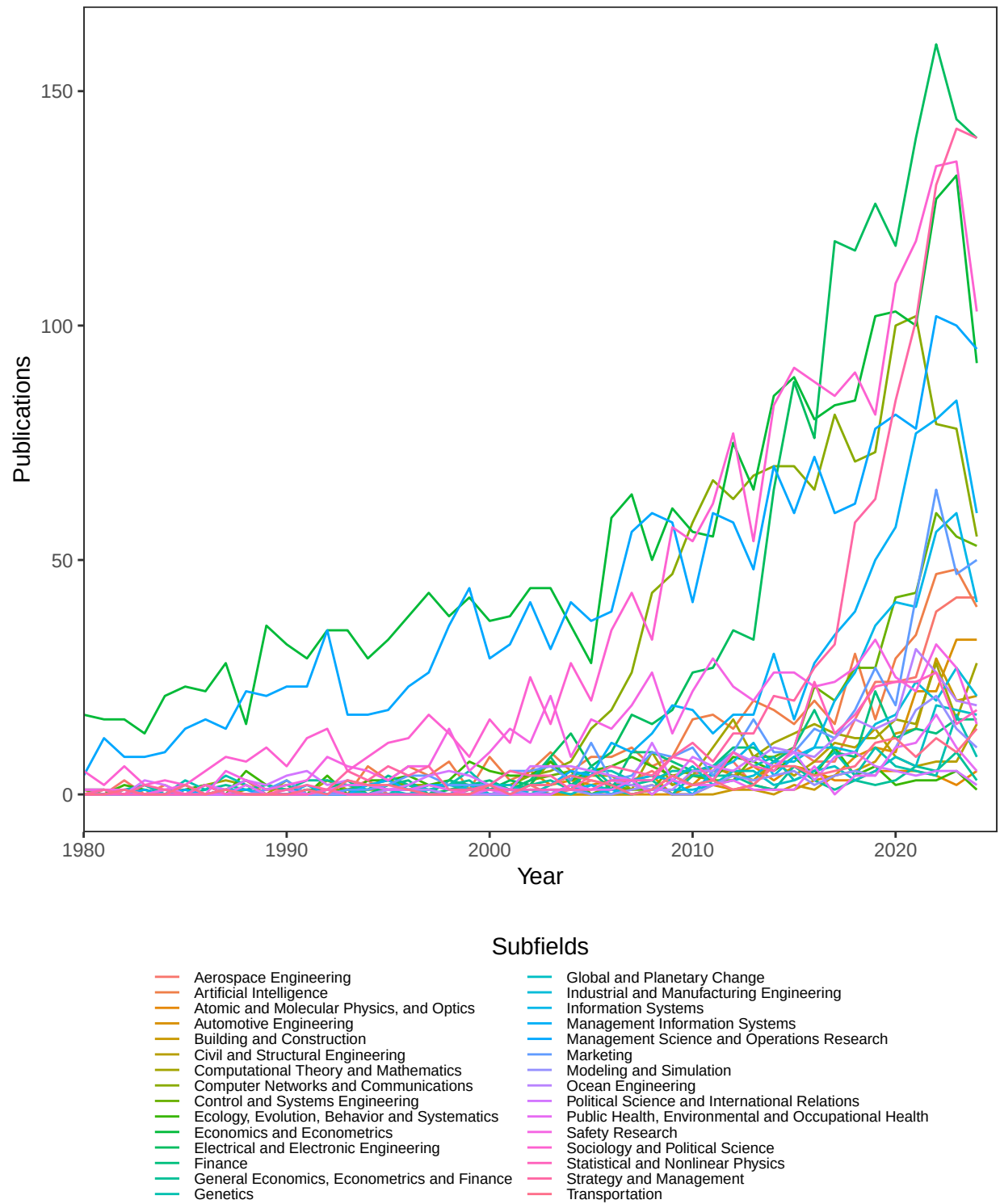
Immunology	5	54
Industrial and Manufacturing Engineering	186	5369
Infectious Diseases	12	240
Information Systems	381	11816
Information Systems and Management	9	155
Insect Science	1	16
Instrumentation	2	34
Language and Linguistics	11	186
Law	11	755
Linguistics and Language	3	166
Literature and Literary Theory	3	26
Management Information Systems	705	22133
Management Science and Operations Research	1879	77654
Management of Technology and Innovation	64	1601
Management, Monitoring, Policy and Law	66	977
Marketing	429	18120
Materials Chemistry	3	363
Mathematical Physics	8	125
Mechanical Engineering	26	424
Mechanics of Materials	11	247
Media Technology	43	828
Microbiology	1	5
Modeling and Simulation	112	2972
Molecular Biology	36	884
Museology	2	2
Music	1	3
Nature and Landscape Conservation	24	1855
Neurology	6	63
Nuclear and High Energy Physics	2	14
Numerical Analysis	12	156
Nutrition and Dietetics	1	0
Ocean Engineering	223	6194
Oceanography	3	144
Oncology	2	28
Organizational Behavior and Human Resource Management	47	1349
Orthopedics and Sports Medicine	3	193
Paleontology	1	32
Parasitology	1	40
Pediatrics, Perinatology and Child Health	2	43
Pharmacology	6	105
Pharmacy	7	65
Philosophy	29	181
Physical Therapy, Sports Therapy and Rehabilitation	1	0
Physiology	3	16
Plant Science	24	718

Political Science and International Relations	183	11694
Pollution	27	658
Polymers and Plastics	1	85
Psychiatry and Mental health	3	15
Public Administration	13	98
Public Health, Environmental and Occupational Health	102	3345
Pulmonary and Respiratory Medicine	7	253
Radiological and Ultrasound Technology	29	651
Radiology, Nuclear Medicine and Imaging	3	58
Renewable Energy, Sustainability and the Environment	74	2654
Safety Research	598	35862
Safety, Risk, Reliability and Quality	45	1449
Sensory Systems	1	16
Signal Processing	24	439
Small Animals	3	54
Social Psychology	44	2326
Sociology and Political Science	1803	77616
Software	3	34
Soil Science	31	2164
Spectroscopy	2	9
Statistical and Nonlinear Physics	251	6656
Statistics and Probability	38	619
Statistics, Probability and Uncertainty	27	687
Strategy and Management	940	35887
Surgery	1	6
Theoretical Computer Science	1	0
Tourism, Leisure and Hospitality Management	3	37
Transportation	134	4854
Urban Studies	11	187
Virology	3	260
Visual Arts and Performing Arts	6	275
Water Science and Technology	23	467

5.2 Top Subfields



5.3 Subfields over Time

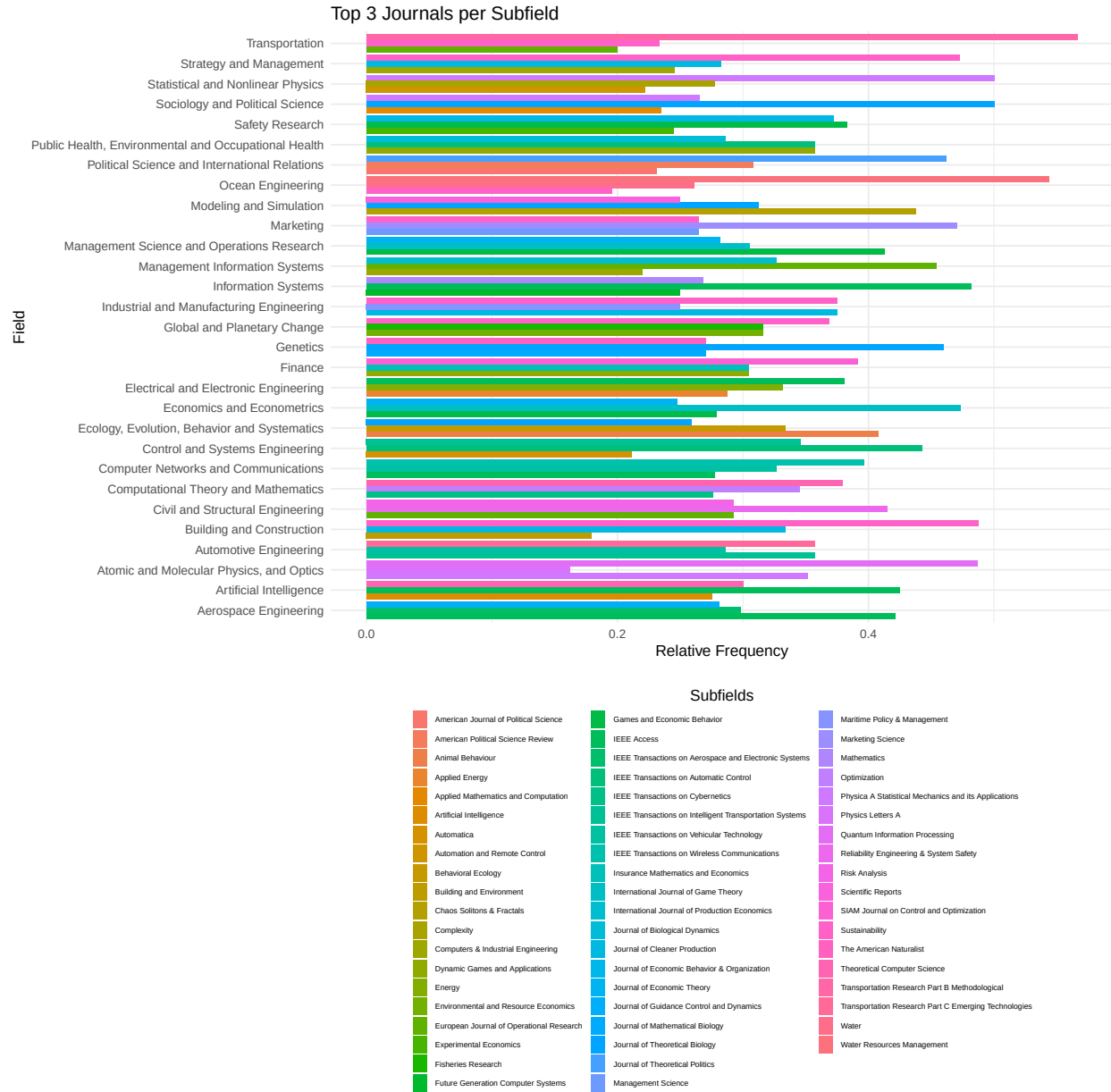


5.4 Distribution of Journals across Subfields

Table 8: Top Journals per Subfield

Subfield	Top Journals per Subfield (Descending Order)		
	Journal 1	Journal 2	Journal 3
Aerospace Engineering	IEEE Access (24)	IEEE Transactions on Aerospace and Electronic Systems (17)	Journal of Guidance Control and Dynamics (16)
Artificial Intelligence	IEEE Access (17)	Theoretical Computer Science (12)	Artificial Intelligence (11)
Atomic and Molecular Physics, and Optics	Quantum Information Processing (18)	Physica A Statistical Mechanics and its Applications (13)	Physics Letters A (6)
Automotive Engineering	IEEE Transactions on Intelligent Transportation Systems (10)	Transportation Research Part C Emerging Technologies (10)	IEEE Transactions on Vehicular Technology (8)
Building and Construction	Sustainability (19)	Journal of Cleaner Production (13)	Building and Environment (7)
Civil and Structural Engineering	Reliability Engineering & System Safety (17)	European Journal of Operational Research (12)	Risk Analysis (12)
Computational Theory and Mathematics	Theoretical Computer Science (11)	Optimization (10)	IEEE Transactions on Cybernetics (8)
Computer Networks and Communications	IEEE Transactions on Wireless Communications (80)	IEEE Transactions on Vehicular Technology (66)	IEEE Access (56)
Control and Systems Engineering	IEEE Transactions on Automatic Control (23)	IEEE Transactions on Intelligent Transportation Systems (18)	Automatica (11)
Ecology, Evolution, Behavior and Systematics	Animal Behaviour (22)	Behavioral Ecology (18)	Journal of Theoretical Biology (14)
Economics and Econometrics	International Journal of Game Theory (107)	Games and Economic Behavior (63)	Journal of Economic Theory (56)
Electrical and Electronic Engineering	IEEE Access (86)	Energy (75)	Applied Energy (65)
Finance	SIAM Journal on Control and Optimization (9)	Dynamic Games and Applications (7)	Insurance Mathematics and Economics (7)
Genetics	Journal of Theoretical Biology (17)	Journal of Mathematical Biology (10)	The American Naturalist (10)
Global and Planetary Change	Sustainability (7)	Environmental and Resource Economics (6)	Fisheries Research (6)
Industrial and Manufacturing Engineering	Journal of Cleaner Production (12)	Sustainability (12)	Maritime Policy & Management (8)
Information Systems	IEEE Access (27)	Mathematics (15)	Future Generation Computer Systems (14)
Management Information Systems	European Journal of Operational Research (64)	International Journal of Production Economics (46)	Computers & Industrial Engineering (31)

Management Science and Operations Research Marketing	Games and Economic Behavior (123) Marketing Science (32)	International Journal of Game Theory (91) Management Science (18)	Journal of Economic Theory (84) Sustainability (18)
Modeling and Simulation	Chaos Solitons & Fractals (7)	Journal of Theoretical Biology (5)	Scientific Reports (4)
Ocean Engineering	Water Resources Management (25)	Water (12)	Sustainability (9)
Political Science and International Relations Public Health, Environmental and Occupational Health Safety Research	Journal of Theoretical Politics (12) Dynamic Games and Applications (5)	American Political Science Review (8) IEEE Transactions on Automatic Control (5)	American Journal of Political Science (6) Journal of Biological Dynamics (4)
	Games and Economic Behavior (36)	Journal of Economic Behavior & Organization (35)	Experimental Economics (23)
Sociology and Political Science	Journal of Theoretical Biology (145)	Physica A Statistical Mechanics and its Applications (77)	Applied Mathematics and Computation (68)
Statistical and Nonlinear Physics	Physica A Statistical Mechanics and its Applications (18)	Complexity (10)	Automation and Remote Control (8)
Strategy and Management Transportation	Sustainability (77)	Journal of Cleaner Production (46)	Computers & Industrial Engineering (40)
	Transportation Research Part B Methodological (17)	Sustainability (7)	European Journal of Operational Research (6)



5.5 Top Articles per Subfield

Table 9: Top Articles per subfield

Top Articles per Subfield (Descending Order)			
Subfield	Authors	Title	No. Citations
Aerospace Engineering	Jingjing Cui;Yuanwei Liu;Arumugam Nallanathan	multi-agent reinforcement learning-based resource allocation for uav networks	498
	Liang Xiao;Xiaozhen Lu;Dongjin Xu;Yuliang Tang;Lei Wang;Weihua Zhuang	uav relay in vanets against smart jamming with reinforcement learning	292
	Xiufeng Song;Peter Willett;Shengli Zhou;Peter B. Luh	the mimo radar and jammer games	183
Artificial Intelligence	Marc Pauly	a modal logic for coalitional power in games	466
	David Poole	the independent choice logic for modelling multiple agents under uncertainty	383
	Zhipeng Cai;Zuobin Xiong;Honghui Xu;Peng Wang;Wei Li;Yi Pan	generative adversarial networks	325
Automotive Engineering	Adam Millard□Ball	pedestrians, autonomous vehicles, and cities	291
	Xiaoxiang Na;David J. Cole	game-theoretic modeling of the steering interaction between a human driver and a vehicle collision avoidance controller	188
	Zhu Xiao;Xingxia Dai;Hongbo Jiang;Dong Wang;Hongyang Chen;Liang Yang;Fanzi Zeng	vehicular task offloading via heat-aware mec cooperation using game-theoretic method	168
Building and Construction	Lei Du;Yingbin Feng;Wei Lu;Lingkai Kong;Zhi Yang	evolutionary game analysis of stakeholders' decision-making behaviours in construction and demolition waste management	148
	Qinghua He;Zilun Wang;Ge Wang;Jian Zuo;Guangdong Wu;Bingsheng Liu	to be green or not to be: how environmental regulations shape contractor greenwashing behaviors in construction projects	131

	Ke Fan;Eddie C.M. Hui	evolutionary game theory analysis for understanding the decision-making mechanisms of governments and developers on green building incentives	126
Civil and Structural Engineering	Jun Zhuang;Vicki M. Bier;Oğuzhan Alagöz	modeling secrecy and deception in a multiple-period attacker–defender signaling game	268
	Vicki M. Bier;Aniruddha Nagaraj;Vinod Abhichandani	protection of simple series and parallel systems with components of different values	204
	Louis Anthony Cox	game theory and risk analysis	162
Computational Theory and Mathematics	Anne Condon	the complexity of stochastic games	501
	Sjur Didrik Flåm;A. C. Антипин	equilibrium programming using proximal-like algorithms	437
	James T. Moore;Jonathan F. Bard	the mixed integer linear bilevel programming problem	343
Computer Networks and Communications	Cem Saraydar;Narayan B. Mandayam;D.J. Goodman	efficient power control via pricing in wireless data networks	1332
	Xu Chen;Xu Chen	decentralized computation offloading game for mobile cloud computing	843
	Yunan Gu;Walid Saad;Mehdi Bennis;Mérouane Debbah;Zhu Han	matching theory for future wireless networks: fundamentals and applications	567
Control and Systems Engineering	Tamer Başar;P. Bernhard	$h/\sup \infty$ -Optimal control and related minimax design problems: a dynamic game approach	1066
	Yang Han;Ke Zhang;Hong Li;E.A.A. Coelho;Josep M. Guerrero	mas-based distributed coordinated control and optimization in microgrid and microgrid clusters: a comprehensive overview	455
	Hao Wang;Jianwei Huang;Hao Wang;Jianwei Huang	incentivizing energy trading for interconnected microgrids	447
Ecology, Evolution, Behavior and Systematics	Mart R. Gross	alternative reproductive strategies and tactics: diversity within sexes	1681

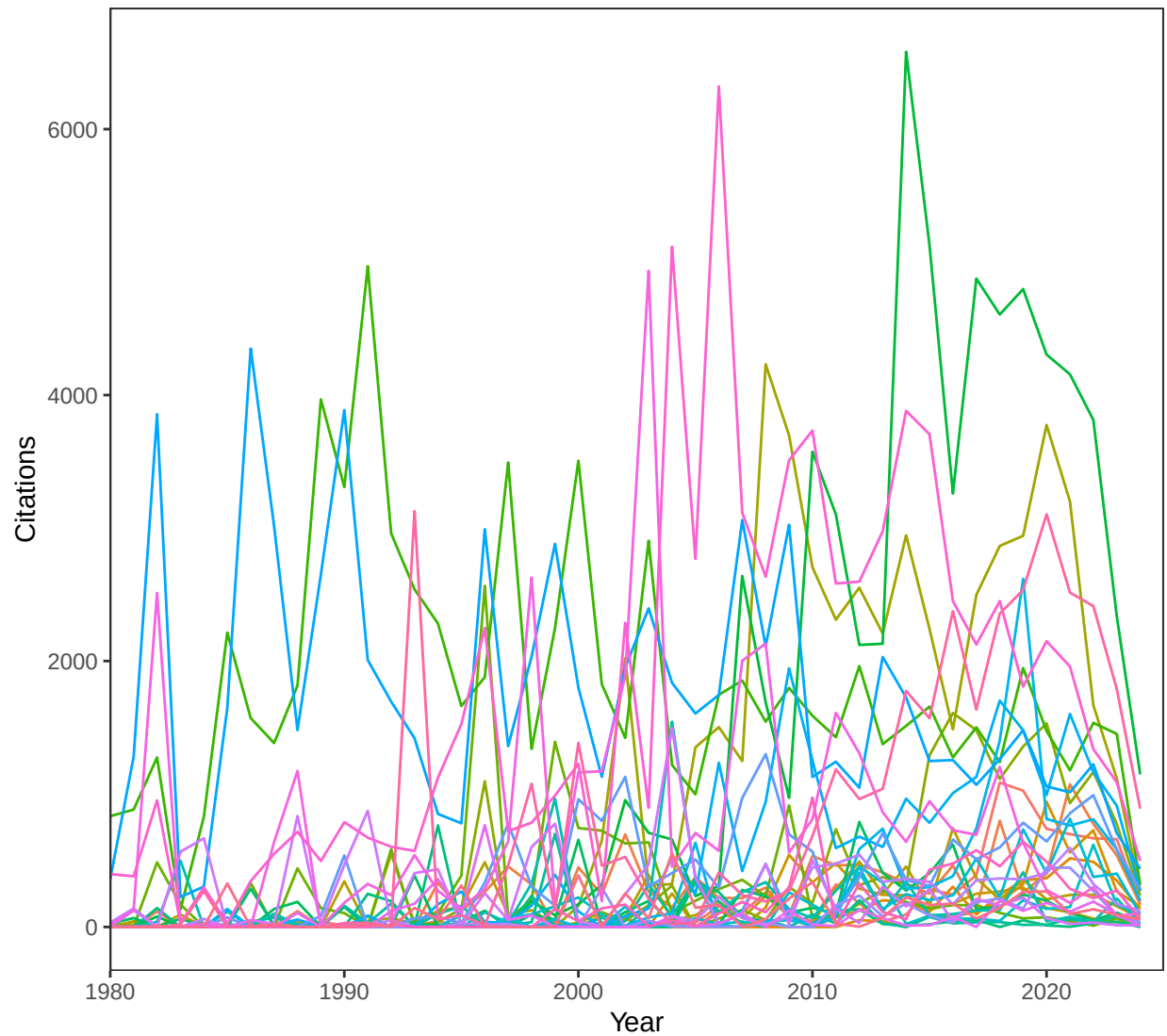
Economics and Econometrics	Hanna Kokko	competition for early arrival in migratory birds	812
	Gareth Arnott;Robert W. Elwood	assessment of fighting ability in animal contests	711
	Charles F. Manski	economic analysis of social interactions	2052
	Daniel Friedman	evolutionary games in economics	1797
	Barry R. Weingast	the political foundations of democracy and the rule of the law	1739
Electrical and Electronic Engineering	Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid	2711
	Amitav Mukherjee;S. Ali. A. Fakoorian;Jing Huang;A. Lee Swindlehurst	principles of physical layer security in multiuser wireless networks: a survey	1427
	Zhongjing Ma;Duncan S. Callaway;Ian A. Hiskens	decentralized charging control of large populations of plug-in electric vehicles	963
	Tomas Björk;Agatha Murgoci;Xun Yu Zhou	mean–variance portfolio optimization with state-dependent risk aversion	508
	Hua Fan;Suresh Sundaresan	debt valuation, renegotiation, and optimal dividend policy	444
Finance	Han Smit;L. A. Ankum	a real options and game-theoretic approach to corporate investment strategy under competition	273
	Colin F. Camerer;Teck-Hua Ho	violations of the betweenness axiom and nonlinearity in probability	740
	Colin F. Camerer;George Loewenstein;Dražen Prelec	neuroeconomics: how neuroscience can inform economics	303
	Teck-Hua Ho;Noah Lim;Colin F. Camerer	modeling the psychology of consumer and firm behavior with behavioral economics	262
	William A. Bomberger;Thomas J. Sargent	the conquest of american inflation	673
General Decision Sciences	Thad Dunning	resource dependence, economic performance, and political stability	265
General Economics, Econometrics and Finance			

Genetics	Francisco J. Ruge-Murciá	inflation targeting under asymmetric preferences	163
	Martin A. Nowak;Karl Sigmund	evolutionary dynamics of biological games	1140
	Paul E. Turner;Lin Chao	prisoner's dilemma in an rna virus	659
Global and Planetary Change	Hans J. Bremermann;John Pickering	a game-theoretical model of parasite virulence	387
	Oskar Franklin;Jacob Johansson;Roderick C. Dewar;Ulf Dieckmann;R. E. McMurtrie;Åke Brännström;Ray Dybzinski	modeling carbon allocation in trees: a search for principles	291
	Jiaqi Peng;Jianmin Zhang	urban flooding risk assessment based on gis- game theory combination weight: a case study of zhengzhou city	169
Industrial and Manufacturing Engineering	Scott Barrett;Astrid Dannenberg	sensitivity of collective action to uncertainty about climate tipping points	168
	Yuandou Wang;Hang Liu;Wanbo Zheng;Yunni Xia;Yawen Li;Peng Chen;Kunyin Guo;Hong Xie	multi-objective workflow scheduling with deep-q-network-based multi-agent reinforcement learning	271
	Masahiro Ishii;Paul Tae-Woo Lee;Koichiro Tezuka;Young-Tae Chang	a game theoretical analysis of port competition	189
Information Systems	Yong Wang;Xiaolei Ma;Zhibin Li;Yong Liu;Maozeng Xu;Yinhai Wang	profit distribution in collaborative multiple centers vehicle routing problem	172
	Wenbo Wang;Dinh Thai Hoang;Peizhao Hu;Zehui Xiong;Dusit Niyato;Ping Wang;Yonggang Wen;Dong In Kim	a survey on consensus mechanisms and mining strategy management in blockchain networks	940
	David Easley;Maureen O'Hara;Soumya Basu	from mining to markets: the evolution of bitcoin transaction fees	547
Management Information Systems	Kong-wei Lye;Jeannette M. Wing	game strategies in network security	433
	Mahesh Nagarajan;Greys Sošić	game-theoretic analysis of cooperation among supply chain agents: review and extensions	607

Management Science and Operations Research	Jan A. Van Mieghem	coordinating investment, production, and subcontracting	361
	Gary H. Chao;Seyed M. R. Iravani;R. Canan Savaskan	quality improvement incentives and product recall cost sharing contracts	358
	David M. Kreps;Robert Wilson	reputation and imperfect information	3101
	Drew Fudenberg;Eric Maskin	the folk theorem in repeated games with discounting or with incomplete information	2371
	Paul Milgrom;John Roberts	rationalizability, learning, and equilibrium in games with strategic complementarities	1816
Marketing	Patrick Bajari;C. Lanier Benkard;Jonathan Levin	estimating dynamic models of imperfect competition	869
	Marcel Corstjens;Rajiv Lal	building store loyalty through store brands	667
	Michael J. Mazzeo	product choice and oligopoly market structure	634
Modeling and Simulation	Christopher A. Klausmeier;Elena Litchman	algal games: the vertical distribution of phytoplankton in poorly mixed water columns	294
	Sheryl L. Chang;Mahendra Piraveenan;Philippa Pattison;Mikhail Prokopenko	game theoretic modelling of infectious disease dynamics and intervention methods: a review	159
	Timothy C. Reluga;Alison P. Galvani	a general approach for population games with application to vaccination	127
	Kaveh Madani;Jay R. Lund	a monte-carlo game theoretic approach for multi-criteria decision making under uncertainty	188
Ocean Engineering	Kaveh Madani;Keith W. Hipel	non-cooperative stability definitions for strategic analysis of generic water resources conflicts	148
	Lu Xiao;Jianyue Liu;Jinwen Ge	dynamic game in agriculture and industry cross-sectoral water pollution governance in developing countries	144
	Alberto Alesina	credibility and policy convergence in a two-party system with rational voters	823
Political Science and International Relations			

Public Health, Environmental and Occupational Health	Wolfgang Mayer	endogenous tariff formation.	666
	Vivien A. Schmidt;Claudio M. Radaelli	policy change and discourse in europe: conceptual and methodological issues	605
	Gerry Mackie	ending footbinding and infibulation: a convention account	765
	Don M. Hugie;Lawrence M. Dill	fish and game: a game theoretic approach to habitat selection by predators and prey*	260
Safety Research	Arja Kaitala;Veijo Kaitala;Per Lundberg	a theory of partial migration	172
	David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	rational cooperation in the finitely repeated prisoners' dilemma	2512
	Martin Dufwenberg;Georg Kirchsteiger	a theory of sequential reciprocity	1902
	Ido Erev;Alvin E. Roth	predicting how people play games: reinforcement learning in experimental games with unique, mixed strategy equilibria	1822
Sociology and Political Science	Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	a simple rule for the evolution of cooperation on graphs and social networks	2155
	Martin A. Nowak;Akira Sasaki;Christine Taylor;Drew Fudenberg	emergence of cooperation and evolutionary stability in finite populations	1378
	Sasha R. X. Dall;Alasdair I. Houston;John M. McNamara	the behavioural ecology of personality: consistent individual differences from an adaptive perspective	1279
	Ramasuri Narayanam;Y. Narahari	a shapley value-based approach to discover influential nodes in social networks	373
Statistical and Nonlinear Physics	Wei Chen;Zhenming Liu;Xiaorui Sun;Yajun Wang	a game-theoretic framework to identify overlapping communities in social networks	217

Strategy and Management	S. Rasoul Etesami;Tamer Başar	game-theoretic analysis of the hegselmann-krause model for opinion dynamics in finite dimensions	176
	Arvind Parkhe	strategic alliance structuring: a game theoretic and transaction cost examination of interfirm cooperation.	2581
	Jan W. Rivkin	imitation of complex strategies	1292
	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	the interorganizational learning dilemma: collective knowledge development in strategic alliances	1077
Transportation	José Correa;Andreas S. Schulz;Nicolás E. Stier-Moses	selfish routing in capacitated networks	386
	Michael G.H. Bell	a game theory approach to measuring the performance reliability of transport networks	379
	Alain Haurie;Patrice Marcotte	on the relationship between nash—cournot and wardrop equilibria	327



Subfields

- | | |
|--|--|
| — Aerospace Engineering | — Global and Planetary Change |
| — Artificial Intelligence | — Industrial and Manufacturing Engineering |
| — Automotive Engineering | — Information Systems |
| — Building and Construction | — Management Information Systems |
| — Civil and Structural Engineering | — Management Science and Operations Research |
| — Computational Theory and Mathematics | — Marketing |
| — Computer Networks and Communications | — Modeling and Simulation |
| — Control and Systems Engineering | — Ocean Engineering |
| — Ecology, Evolution, Behavior and Systematics | — Political Science and International Relations |
| — Economics and Econometrics | — Public Health, Environmental and Occupational Health |
| — Electrical and Electronic Engineering | — Safety Research |
| — Finance | — Sociology and Political Science |
| — General Decision Sciences | — Statistical and Nonlinear Physics |
| — General Economics, Econometrics and Finance | — Strategy and Management |
| — Genetics | — Transportation |

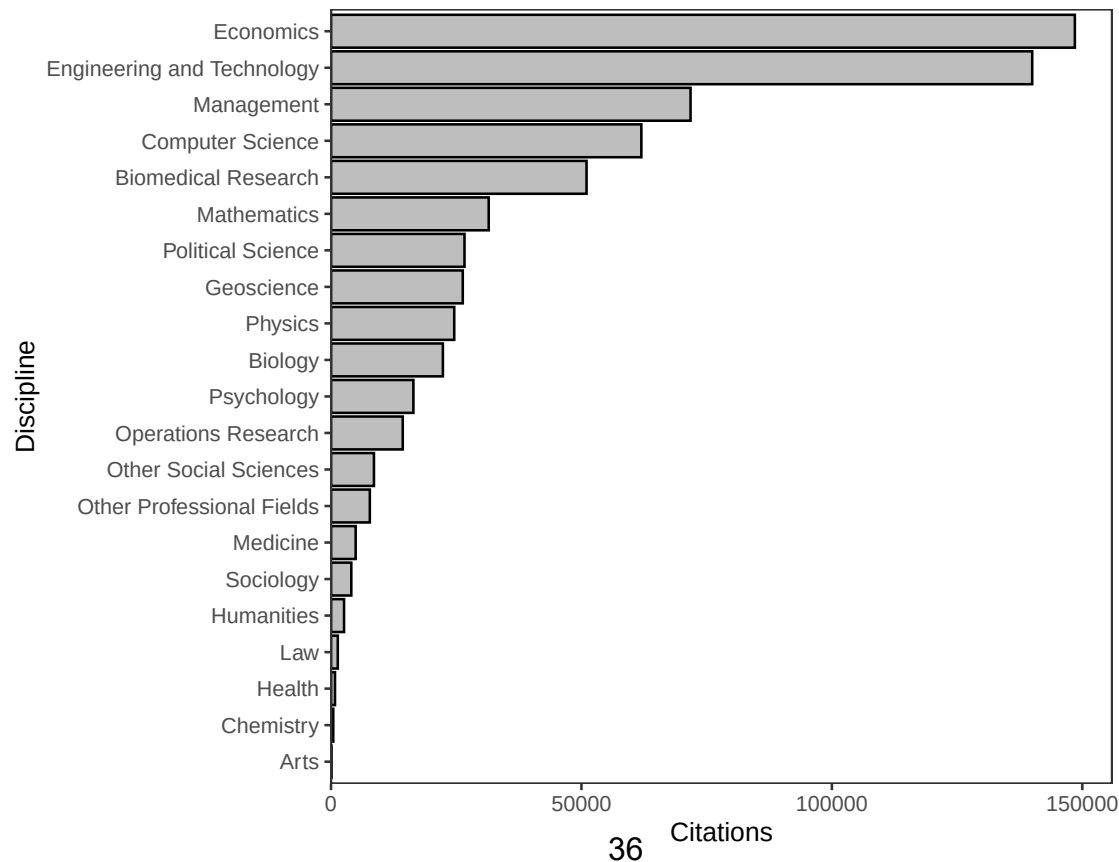
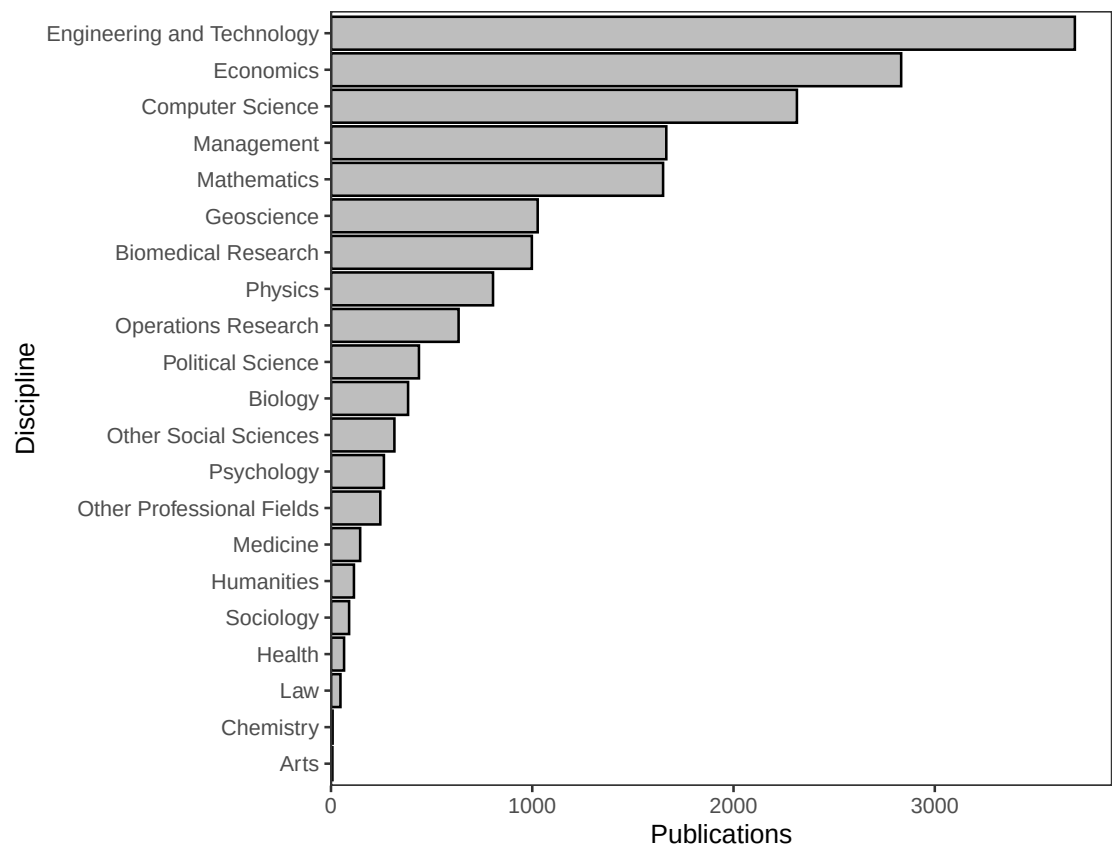
6. Discipline Overview

6.1 List of Disciplines

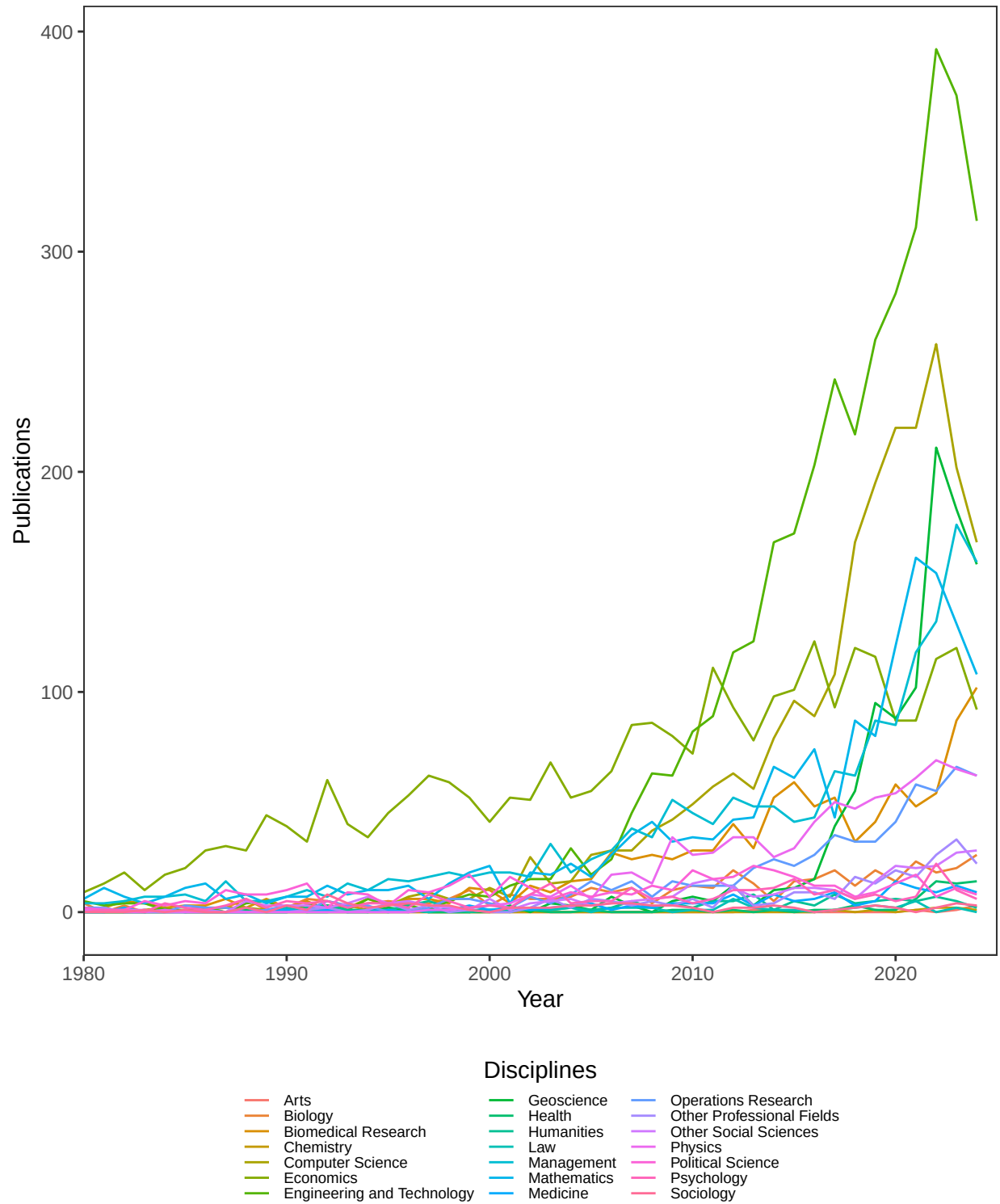
Table 10: List of Disciplines

Discipline	No. Publications	No. Citations
Arts	7	56
Biology	383	22338
Biomedical Research	998	51021
Chemistry	8	452
Computer Science	2315	61960
Economics	2833	148510
Engineering and Technology	3696	139995
Geoscience	1027	26311
Health	65	802
Humanities	114	2607
Law	47	1358
Management	1666	71784
Mathematics	1650	31491
Medicine	145	4924
Operations Research	634	14327
Other Professional Fields	245	7754
Other Social Sciences	315	8577
Physics	805	24603
Political Science	437	26655
Psychology	263	16448
Sociology	90	4054

6.2 Top Disciplines



6.3 Disciplines over Time



6.4 Distribution of Journals across Disciplines

Table 11: Top Journals per Discipline

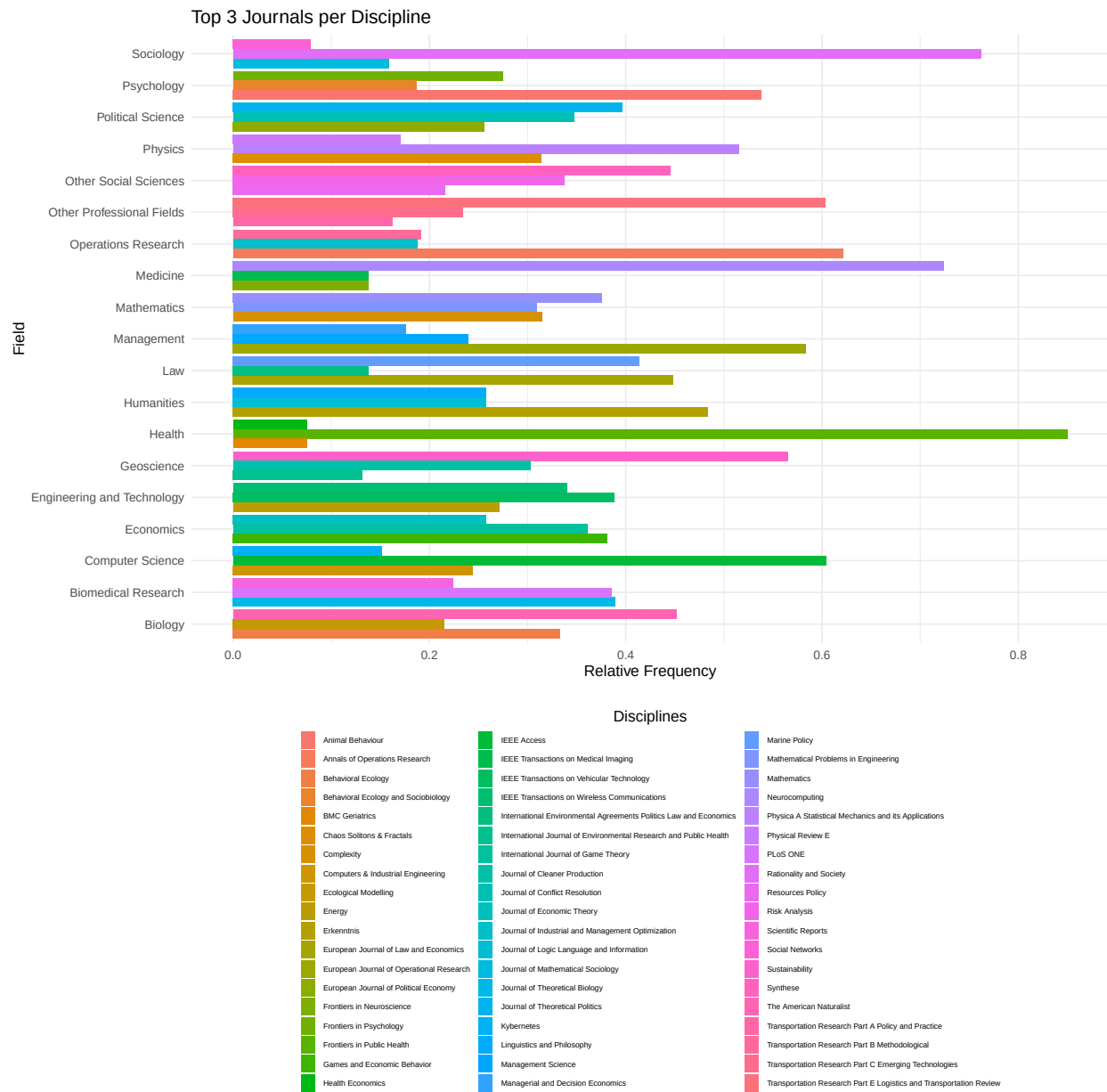
Discipline	Top Journals per discipline (Descending Order)		
	Journal 1	Journal 2	Journal 3
Biology	The American Naturalist (42)	Behavioral Ecology (31)	Ecological Modelling (20)
Biomedical Research	Journal of Theoretical Biology (210)	PLoS ONE (208)	Scientific Reports (121)
Computer Science	IEEE Access (346)	Computers & Industrial Engineering (140)	Kybernetes (87)
Economics	Games and Economic Behavior (241)	International Journal of Game Theory (228)	Journal of Economic Theory (163)
Engineering and Technology	IEEE Transactions on Vehicular Technology (169)	IEEE Transactions on Wireless Communications (148)	Energy (118)
Geoscience	Sustainability (360)	Journal of Cleaner Production (193)	International Journal of Environmental Research and Public Health (84)
Health	Frontiers in Public Health (34)	BMC Geriatrics (3)	Health Economics (3)
Humanities	Erkenntnis (15)	Journal of Logic Language and Information (8)	Linguistics and Philosophy (8)
Law	European Journal of Law and Economics (13)	Marine Policy (12)	International Environmental Agreements Politics Law and Economics (4)
Management	European Journal of Operational Research (275)	Management Science (113)	Managerial and Decision Economics (83)
Mathematics	Mathematics (141)	Complexity (118)	Mathematical Problems in Engineering (116)
Medicine	Neurocomputing (21)	Frontiers in Neuroscience (4)	IEEE Transactions on Medical Imaging (4)
Operations Research	Annals of Operations Research (169)	Transportation Research Part B Methodological (52)	Journal of Industrial and Management Optimization (51)
Other Professional Fields	Transportation Research Part E Logistics and Transportation Review (67)	Transportation Research Part C Emerging Technologies (26)	Transportation Research Part A Policy and Practice (18)
Other Social Sciences	Synthese (33)	Risk Analysis (25)	Resources Policy (16)
Physics	Physica A Statistical Mechanics and its Applications (151)	Chaos Solitons & Fractals (92)	Physical Review E (50)
Political Science	Journal of Theoretical Politics (48)	Journal of Conflict Resolution (42)	European Journal of Political Economy (31)
Psychology	Animal Behaviour (49)	Frontiers in Psychology (25)	Behavioral Ecology and Sociobiology (17)

Sociology

Rationality and Society
(48)

Journal of Mathematical
Sociology (10)

Social Networks (5)



6.5 Top Articles per Discipline

Table 12: Top Articles per Discipline

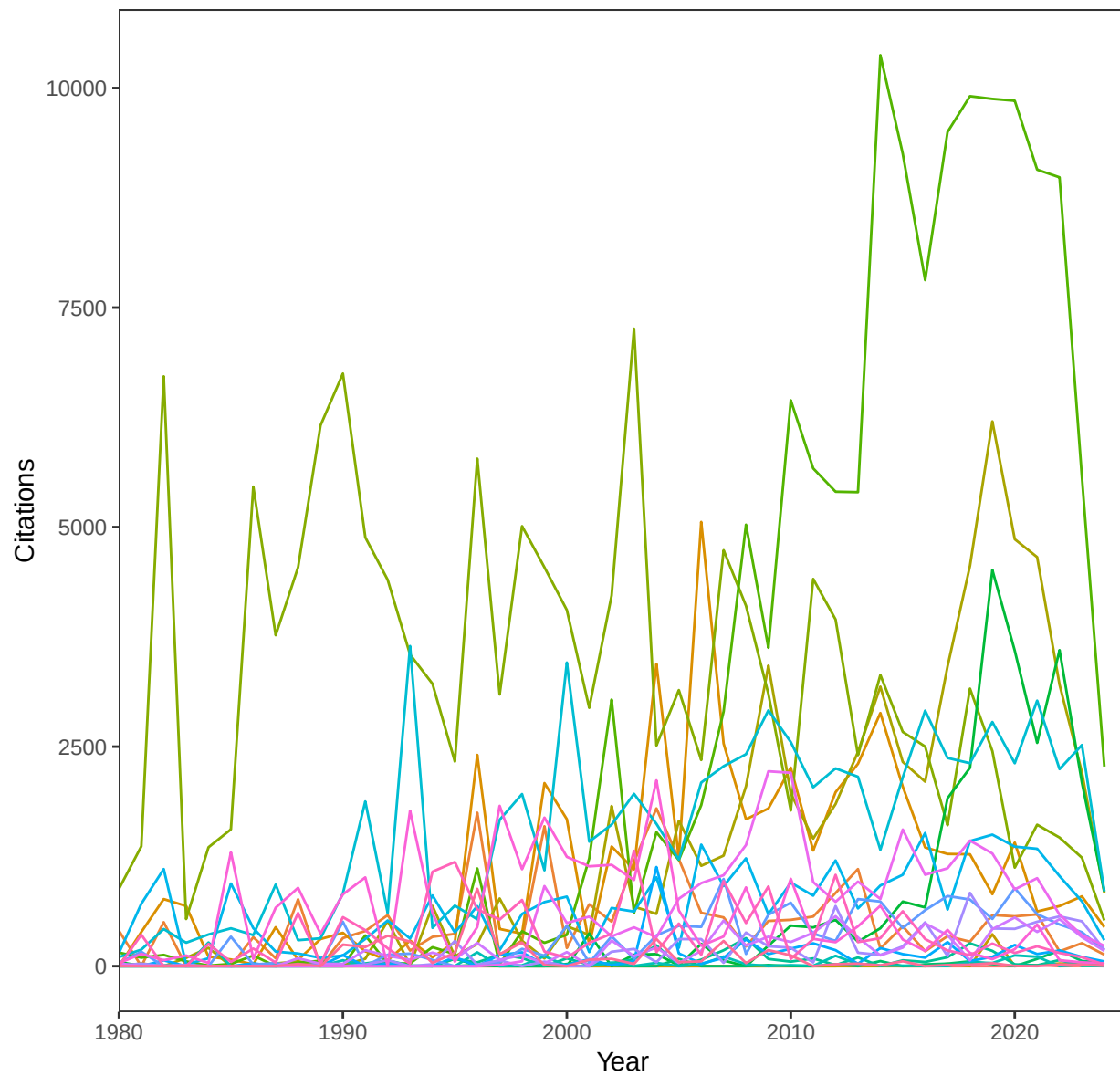
Top Articles per Discipline (Descending Order)			
Discipline	Authors	Title	No. Citations
Arts	Y.G. Li;Ruiyuan Cong;Keyong Zhang;Shenglin Ma;Chunli Fu	four-way game analysis of transformation and upgrading of manufacturing enterprises relying on industrial internet platform under developers' participation	17
	Clément L. Canonne	focal points in collective free improvisation	14
	Shaowen Wang;Xiaojun Liu	development of epc model in chinese public projects: evolutionary game among stakeholders	13
Biology	Mart R. Gross	alternative reproductive strategies and tactics: diversity within sexes	1681
	Sasha R. X. Dall;Alasdair I. Houston;John M. McNamara	the behavioural ecology of personality: consistent individual differences from an adaptive perspective	1279
	Michael Doebeli;Christoph Hauert	models of cooperation based on the prisoner's dilemma and the snowdrift game	861
Biomedical Research	Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	a simple rule for the evolution of cooperation on graphs and social networks	2155
	Martin A. Nowak;Akira Sasaki;Christine Taylor;Drew Fudenberg	emergence of cooperation and evolutionary stability in finite populations	1378
	Matjaž Perc;Jesus Gomez- Gardenes;Attila Szolnoki;Luis M. Floria;Yamir Moreno	evolutionary dynamics of group interactions on structured populations: a review	1247
Chemistry	Han Yue Zhang;Yuan Yuan Tang;Ping Ping Shi;Ren Gen Xiong	toward the targeted design of molecular ferroelectrics: modifying molecular symmetries and homochirality	363

Computer Science	Suzanne R. McAnanama-Brereton; Mark P. Waller	rational density functional selection using game theory	27
	Arthur Cartel Foahom Gouabou; Jules Collenne; Jilliana Monnier; Rabah Iguernaissi; Jean-Luc Damoiseaux; Abdellatif Moudafi; Djamal Merad	computer aided diagnosis of melanoma using deep neural networks and game theory: application on dermoscopic images of skin lesions	21
	Wenbo Wang; Dinh Thai Hoang; Peizhao Hu; Zehui Xiong; Dusit Niyato; Ping Wang; Yonggang Wen; Dong In Kim	a survey on consensus mechanisms and mining strategy management in blockchain networks	940
	Constantinos Daskalakis; Paul W. Goldberg; Christos H. Papadimitriou	the complexity of computing a nash equilibrium	921
	Xu Chen; Xu Chen	decentralized computation offloading game for mobile cloud computing	843
Economics	David M. Kreps; Robert Wilson	reputation and imperfect information	3101
	David M. Kreps; Paul Milgrom; John Roberts; Robert Wilson	rational cooperation in the finitely repeated prisoners' dilemma	2512
	Drew Fudenberg; Eric Maskin	the folk theorem in repeated games with discounting or with incomplete information	2371
Engineering and Technology	Amir-Hamed Mohsenian-Rad; Vincent W. S. Wong; Juri Jatskevich; Robert Schober; Alberto Leon-Garcia	autonomous demand-side management based on game-theoretic energy consumption scheduling for the future smart grid	2711
	Amitav Mukherjee; S. Ali. A. Fakoorian; Jing Huang; A. Lee Swindlehurst	principles of physical layer security in multiuser wireless networks: a survey	1427
	Cem Saraydar; Narayan B. Mandayam; D.J. Goodman	efficient power control via pricing in wireless data networks	1332
Geoscience	Wanting Chen; Zhi-Hua Hu	using evolutionary game theory to study governments and manufacturers' behavioral strategies under various carbon taxes and subsidies	371

Health	Juhong Gao;Hongshuai Han;Liting Hou;Haiyan Wang	pricing and effort decisions in a closed-loop supply chain under different channel power structures	314
	Christopher A. Klausmeier;Elena Litchman	algal games: the vertical distribution of phytoplankton in poorly mixed water columns	294
	Amori Yee Mikami;Margaret A. Boucher;Keith Humphreys	prevention of peer rejection through a classroom-level intervention in middle school	84
	Kris De Jaegher;Marc Jegers	the physician–patient relationship as a game of strategic information transmission	66
	Bradley Chen;Victoria Y. Fan	strategic provider behavior under global budget payment with price adjustment in taiwan	58
Humanities	Bryce Huebner	norms in the wild: how to diagnose, measure, and change social norms	225
	Robert van Rooy	signalling games select horn strategies	183
	Mark G. Chen	communication, coordination, and camaraderie in world of warcraft	164
Law	John T. Scholz	cooperation, deterrence, and the ecology of regulatory enforcement	270
	Jack Knight;Lee Epstein	on the struggle for judicial supremacy	158
	U. Rashid Sumaila	a review of game-theoretic models of fishing	113
Management	Arvind Parkhe	strategic alliance structuring: a game theoretic and transaction cost examination of interfirm cooperation.	2581
	Jan W. Rivkin	imitation of complex strategies	1292
	Rikard Larsson;Lars Bengtsson;Kristina Henriksson;Judith Sparks	the interorganizational learning dilemma: collective knowledge development in strategic alliances	1077

Mathematics	Alvin E. Roth	the economics of matching: stability and incentives	895
	Xiaotie Deng;Christos H. Papadimitriou	on the complexity of cooperative solution concepts	613
	Paul Milgrom;Robert J. Weber	distributional strategies for games with incomplete information	568
Medicine	Dominic J. Barraclough;Michelle Conroy;Daeyeol Lee	prefrontal cortex and decision making in a mixed-strategy game	696
	Michael C. Dorris;Paul W. Glimcher	activity in posterior parietal cortex is correlated with the relative subjective desirability of action	426
	Daeyeol Lee	game theory and neural basis of social decision making	304
Operations Research	Jun Zhuang;Vicki M. Bier	balancing terrorism and natural disasters—defensive strategy with endogenous attacker effort	422
	Michael G.H. Bell	a game theory approach to measuring the performance reliability of transport networks	379
	James T. Moore;Jonathan F. Bard	the mixed integer linear bilevel programming problem	343
Other Professional Fields	Huseyin Cavusoglu;Srinivasan Raghunathan;Wei Thoo Yue	decision-theoretic and game-theoretic approaches to it security investment	225
	Xu Chen;Xiaojun Wang;Hing Kai Chan	manufacturer and retailer coordination for environmental and economic competitiveness: a power perspective	206
	Masahiro Ishii;Paul Tae-Woo Lee;Koichiro Tezuka;Young-Tae Chang	a game theoretical analysis of port competition	189
Other Social Sciences	Adam Millard-JBall	pedestrians, autonomous vehicles, and cities	291
	Magnus Enquist;Kimmo Eriksson;Stefano Ghirlanda	critical social learning: a solution to rogers's paradox of nonadaptive culture	244

Physics	Eddie C.M. Hui;Haijun Bao	the logic behind conflicts in land acquisitions in contemporary china: a framework based upon game theory	242
	David Meyer	quantum strategies	744
	Zhen Wang;Lin Wang;Attila Szolnoki;Matjaž Perc;Zhen Wang;Lin Wang;Attila Szolnoki;Matjaž Perc	evolutionary games on multilayer networks: a colloquium	703
Political Science	Christoph Hauert;György Szabó	game theory and physics	490
	Barry R. Weingast	the political foundations of democracy and the rule of the law	1739
	John H. Aldrich	rational choice and turnout	1210
Psychology	David Austen-Smith;Jeffrey S. Banks	elections, coalitions, and legislative outcomes	752
	Linda Mealey	the sociobiology of sociopathy: an integrated evolutionary model	984
	Ronald Nozick;Peter Hammerstein	biological markets: supply and demand determine the effect of partner choice in cooperation, mutualism and mating	882
Sociology	Gareth Arnott;Robert W. Elwood	assessment of fighting ability in animal contests	711
	Gerry Mackie	ending footbinding and infibulation: a convention account	765
	Junfu Zhang	a dynamic model of residential segregation	229
	Michael W. Macy	learning to cooperate: stochastic and tacit collusion in social exchange	218



Disciplines

- | | |
|----------------------------|---------------------------|
| Arts | Management |
| Biology | Mathematics |
| Biomedical Research | Medicine |
| Chemistry | Operations Research |
| Computer Science | Other Professional Fields |
| Economics | Other Social Sciences |
| Engineering and Technology | Physics |
| Geoscience | Political Science |
| Health | Psychology |
| Humanities | Sociology |
| Law | |

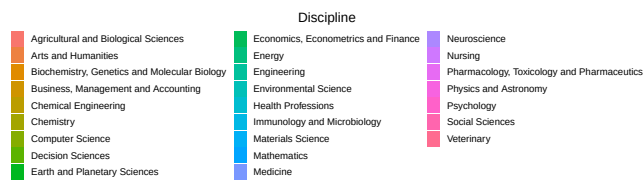
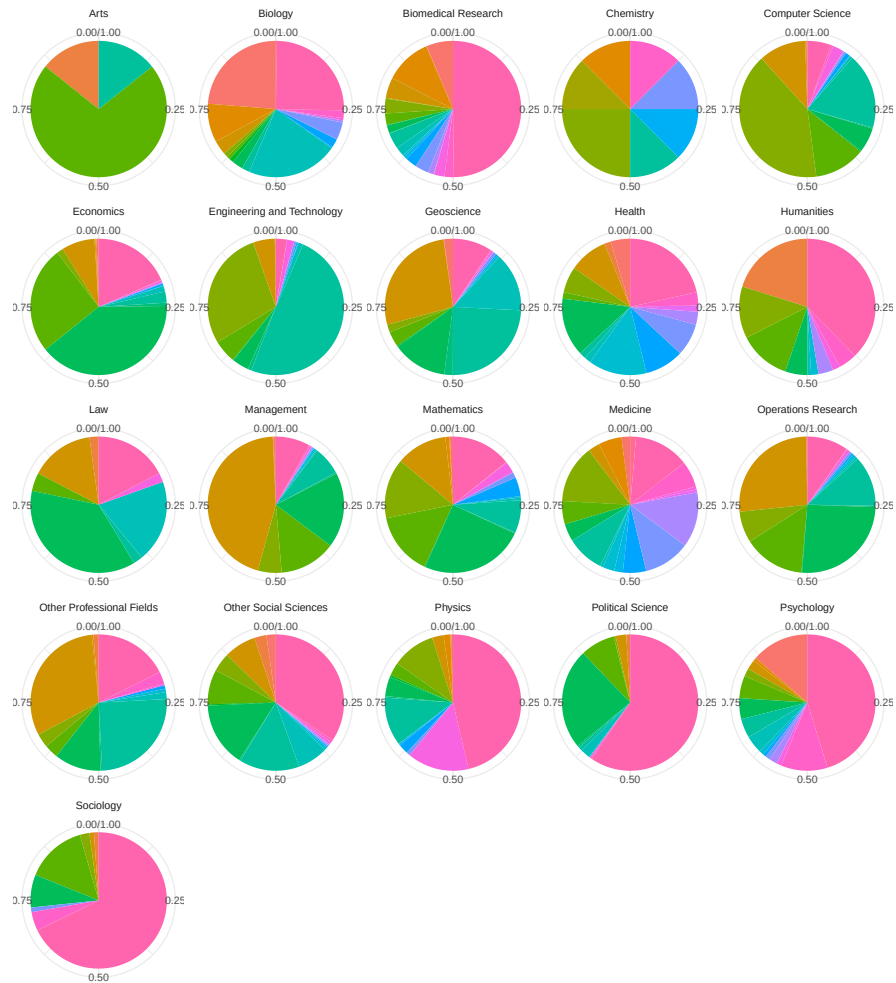
6.6 Discipline Composition of Field (by OA)

Discipline Composition of Fields



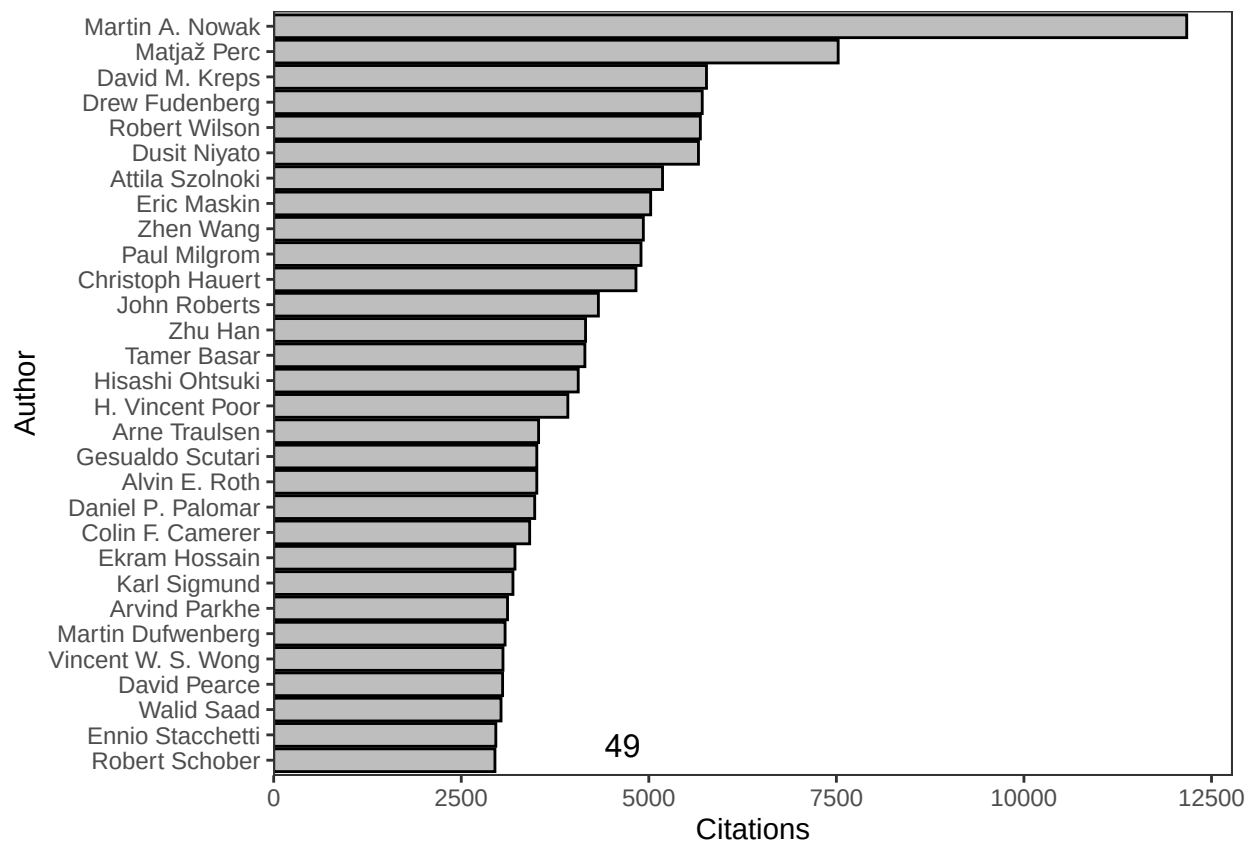
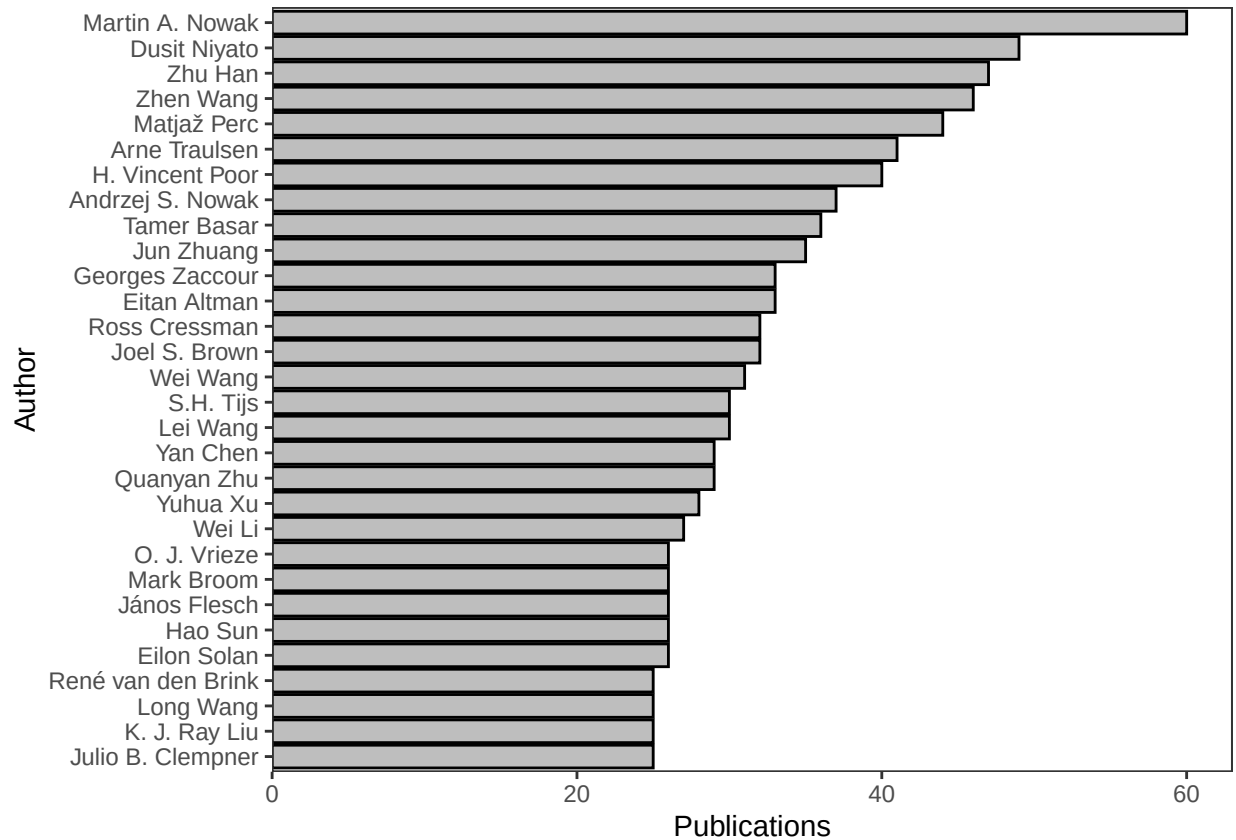
6.7 Field Composition (by OA) of Disciplines

Discipline Composition of Fields



7. Authors Overview

7.1 Top Authors



7.2 Top Authors per Domain

Table 13: Top Authors in Terms of Citations per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
Gerry Mackie (765)	Mart R. Gross (1681)	Dusit Niyato (5662)	Martin A. Nowak (10999)
Don M. Hugie (260)	Joel S. Brown (1384)	Zhu Han (4031)	Matjaž Perc (7279)
Lawrence M. Dill (260)	Hanna Kokko (1265)	Tamer Başar (3647)	David M. Kreps (5769)
Alison P. Galvani (245)	Rufus A. Johnstone (1262)	H. Vincent Poor (3560)	Drew Fudenberg (5712)
Eunha Shim (245)	Karl Sigmund (1194)	Gesualdo Scutari (3497)	Robert Wilson (5687)

Table 14: Top Authors in Terms of Publications per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
K. M. Ariful Kabir (5)	Joel S. Brown (15)	Dusit Niyato (49)	Martin A. Nowak (56)
Paolo Russu (5)	Ross Cressman (9)	Zhu Han (43)	Matjaž Perc (39)
Angelo Antoci (4)	Jan Rychtář (8)	H. Vincent Poor (36)	Arne Traulsen (38)
Jan Rychtář (4)	J. M. Cushing (7)	Tamer Başar (26)	Andrzej S. Nowak (36)
Monica–Gabriela Cojocar (4)	Gordon G. McNickle (6)	Yuhua Xu (26)	Zhen Wang (33)

7.3 Top Authors per Field

Table 15: Top Authors in Terms of Citations per Field

Field	Authors		
Business, Management and Accounting	Arvind Parkhe (3117)	Rajiv Lal (1604)	Debabrata Ghosh (1399)
Computer Science	Dusit Niyato (4370)	Ekram Hossain (2180)	Narayan B. Mandayam (2170)
Decision Sciences	Robert Wilson (3153)	Eric Maskin (3152)	David M. Kreps (3101)
Economics, Econometrics and Finance	Charles F. Manski (2052)	Dilip Abreu (2039)	David Pearce (2005)
Engineering	Vincent W. S. Wong (3013)	Robert Schober (2916)	Daniel P. Palomar (2715)
Mathematics	Christopher A. Klausmeier (294)	Elena Litchman (294)	Donald A. Martin (259)
Physics and Astronomy	David Meyer (744)	Hui-Jia Li (503)	Jie Cao (427)
Psychology	Herbert Gintis (411)	Geoffrey F. Miller (399)	Peter M. Todd (368)
Social Sciences	Martin A. Nowak (10842)	Matjaž Perc (7273)	Attila Szolnoki (5183)

Table 16: Top Authors in Terms of Publications per Field

Field	Authors		
Business, Management and Accounting	Tsan-Ming Choi (15)	Ata Allah Taleizadeh (14)	Junhai Ma (14)
Computer Science	Dusit Niyato (27)	Zhu Han (19)	H. Vincent Poor (18)
Decision Sciences	Ehud Lehrer (15)	Eilon Solan (15)	János Flesch (15)
Economics, Econometrics and Finance	Andrzej S. Nowak (35)	S.H. Tijs (24)	René van den Brink (23)
Engineering	Zhu Han (22)	Dusit Niyato (21)	Walid Saad (17)
Mathematics	Jun Tanimoto (4)	K. M. Ariful Kabir (4)	Marco Archetti (4)
Physics and Astronomy	Azhar Iqbal (7)	Derek Abbott (7)	Hui-Jia Li (7)
Psychology	Alasdair I. Houston (2)	David R. Ewoldsen (2)	Dennis R. Brophy (2)
Social Sciences	Martin A. Nowak (55)	Arne Traulsen (38)	Matjaž Perc (38)

7.4 Top Authors per Subfield

Table 17: Top Authors in Terms of Citations per Subfield

Subfield	Authors		
Aerospace Engineering	Arumugam Nallanathan (498)	Jingjing Cui (498)	Yuanwei Liu (498)
Artificial Intelligence	Changchuan Yin (499)	Mingzhe Chen (499)	Walid Saad (499)
Automotive Engineering	Xiaoxiang Na (455)	David J. Cole (310)	Adam Millard □ Ball (291)
Building and Construction	Zhi Yang (275)	Jian Zuo (234)	Guangdong Wu (214)
Civil and Structural Engineering	Jun Zhuang (880)	Vicki M. Bier (597)	Oğuzhan Alagöz (268)
Computational Theory and Mathematics	Frank L. Lewis (611)	Anne Condon (501)	Kyriakos G. Vamvoudakis (268)
Computer Networks and Communications	Dusit Niyato (2671)	Ekram Hossain (2180)	Narayan B. Mandayam (268)
Control and Systems Engineering	Tamer Başar (1722)	P. Bernhard (1066)	Hao Wang (894)
Ecology, Evolution, Behavior and Systematics	Mart R. Gross (1681)	Hanna Kokko (1265)	Rufus A. Johnstone (1265)
Economics and Econometrics	Charles F. Manski (2052)	Dilip Abreu (2039)	David Pearce (2005)
Electrical and Electronic Engineering	Vincent W. S. Wong (3013)	Robert Schober (2916)	Daniel P. Palomar (2715)
Finance	Agatha Murgoci (728)	Tomas Björk (728)	Suresh Sundaresan (559)
General Decision Sciences	Colin F. Camerer (1305)	Teck □ Hua Ho (1002)	Dražen Prelec (303)
General Economics, Econometrics and Finance	Thomas J. Sargent (673)	William A. Bomberger (673)	Francisco J. Ruge-Murcia (303)
Genetics	Karl Sigmund (1194)	Martin A. Nowak (1162)	Lin Chao (795)
Global and Planetary Change	Ray Dybzinski (426)	Jacob Johansson (291)	Oskar Franklin (291)
Industrial and Manufacturing Engineering	Yong Wang (318)	Maozeng Xu (301)	Xiaolei Ma (301)
Information Systems	Dusit Niyato (1457)	Ping Wang (1396)	Wenbo Wang (1396)
Management Information Systems	Greys Sošić (607)	Mahesh Nagarajan (607)	Tiaojun Xiao (576)
Management Science and Operations Research	Robert Wilson (3153)	Eric Maskin (3152)	David M. Kreps (3101)
Marketing	Rajiv Lal (1604)	C. Lanier Benkard (869)	Jonathan Levin (869)
Modeling and Simulation	Christopher A. Klausmeier (294)	Elena Litchman (294)	Jun Tanimoto (255)
Ocean Engineering	Kaveh Madani (556)	Reza Kerachian (453)	Armaghan Abed □ Elmdoudi (255)
Political Science and International Relations	Alberto Alesina (823)	Wolfgang Mayer (666)	Robert Powell (653)
Public Health, Environmental and Occupational Health	Gerry Mackie (765)	Don M. Hugie (260)	Lawrence M. Dill (260)
Safety Research	Martin Dufwenberg (2977)	David M. Kreps (2512)	John Roberts (2512)
Sociology and Political Science	Martin A. Nowak (10593)	Matjaž Perc (7209)	Attila Szolnoki (5177)
Statistical and Nonlinear Physics	Hui □ Jia Li (503)	Jie Cao (427)	Zhan Bu (420)
Strategy and Management	Arvind Parkhe (3117)	Debabrata Ghosh (1392)	Janat Shah (1392)
Transportation	Michael G.H. Bell (593)	Nicolás E. Stier □ Moses (412)	Andreas S. Schulz (386)

Table 18: Top Authors in Terms of Publications per Subfield

Subfield	Authors		
Aerospace Engineering	Chenguang Shi (9)	Jianjiang Zhou (9)	I. S. Rappoport (6)
Artificial Intelligence	Michael Wooldridge (8)	Zhen Zhang (6)	Dongqing Wang (5)
Atomic and Molecular Physics, and Optics	Azhar Iqbal (6)	Derek Abbott (6)	Edward W. Piotrowski (5)
Automotive Engineering	Xiqun Chen (5)	Dong Mo (4)	Shaobo Lu (4)
Building and Construction	Feng Xiong (3)	Qingfeng Meng (3)	Shuaian Wang (3)
Civil and Structural Engineering	Jun Zhuang (10)	Julio B. Clempner (6)	Genserik Reniers (5)
Computational Theory and Mathematics	Frank L. Lewis (5)	Huanguang Zhang (5)	I. V. Konnov (5)
Computer Networks and Communications	Dusit Niyato (18)	H. Vincent Poor (17)	Zhu Han (17)
Control and Systems Engineering	Tamer Başar (8)	Han Zhao (7)	Daniel E. Quevedo (6)
Ecology, Evolution, Behavior and Systematics	Hanna Kokko (5)	Leigh W. Simmons (5)	Luc-Alain Giraldeau (5)
Economics and Econometrics	Andrzej S. Nowak (35)	René van den Brink (23)	S.H. Tijs (23)
Electrical and Electronic Engineering	Zhu Han (21)	Dusit Niyato (19)	H. Vincent Poor (14)
Finance	Han Smit (5)	Sheng Li (5)	Chandan Pal (4)
General Economics, Econometrics and Finance	Assar Lindbeck (2)	Francisco J. Ruge-Murciá (2)	Jan Libich (2)
Genetics	Joel S. Brown (10)	Ross Cressman (9)	J. M. Cushing (6)
Global and Planetary Change	U. Rashid Sumaila (5)	Gordon R. Munro (3)	Marko Lindroos (3)
Industrial and Manufacturing Engineering	Meifeng Luo (4)	Yong Wang (4)	George Q. Huang (3)
Information Systems	Xing Gao (7)	Dusit Niyato (5)	Quanyan Zhu (5)
Management Information Systems	Tiaojun Xiao (10)	Ata Allah Taleizadeh (9)	Junhai Ma (9)
Management Science and Operations Research	Ehud Lehrer (15)	Eilon Solan (15)	János Flesch (15)
Marketing	Salma Karray (11)	Guíomar Martín-Herrán (8)	Ruiliang Yan (8)
Modeling and Simulation	Jun Tanimoto (4)	K. M. Ariful Kabir (4)	Marco Archetti (4)
Ocean Engineering	Reza Kerachian (12)	Armaghan Abed-Elmdoust (6)	Richárd Kicsiny (5)
Political Science and International Relations	George Tsebelis (5)	Christophe Crombez (4)	David Austen-Smith (2)
Public Health, Environmental and Occupational Health	K. M. Ariful Kabir (4)	Monica-Gabriela Cojocaru (4)	Mark Broom (3)
Safety Research	Colin F. Camerer (8)	Amnon Rapoport (7)	Andrew M. Colman (7)
Sociology and Political Science	Martin A. Nowak (53)	Arne Traulsen (38)	Matjaž Perc (37)
Statistical and Nonlinear Physics	Hui-Jia Li (7)	Jie Cao (7)	Yan Chen (6)
Strategy and Management	Ashkan Hafezalkotob (8)	Wei Li (7)	Kannan Govindan (6)
Transportation	Byung-Wook Wie (5)	Chengguang Liu (3)	Fenling Feng (3)

7.5 Top Authors per Discipline

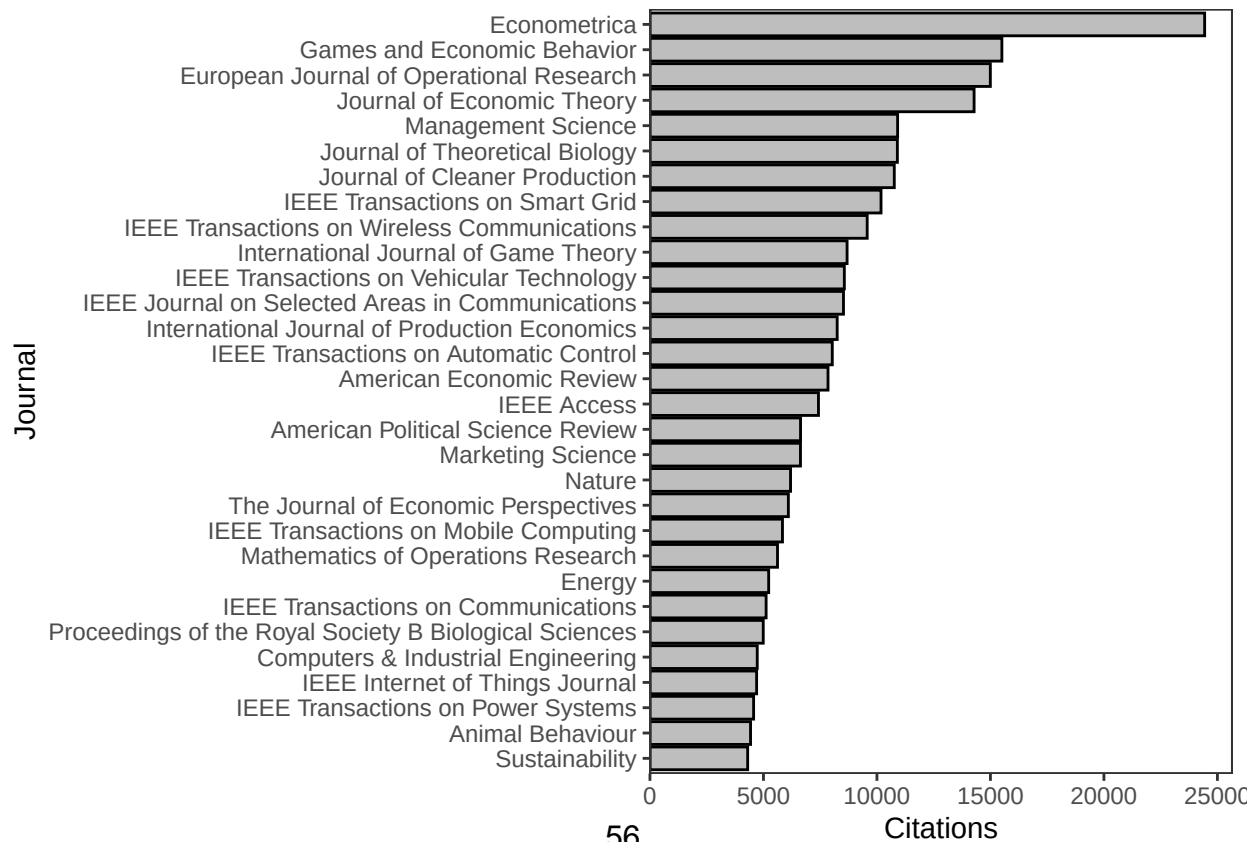
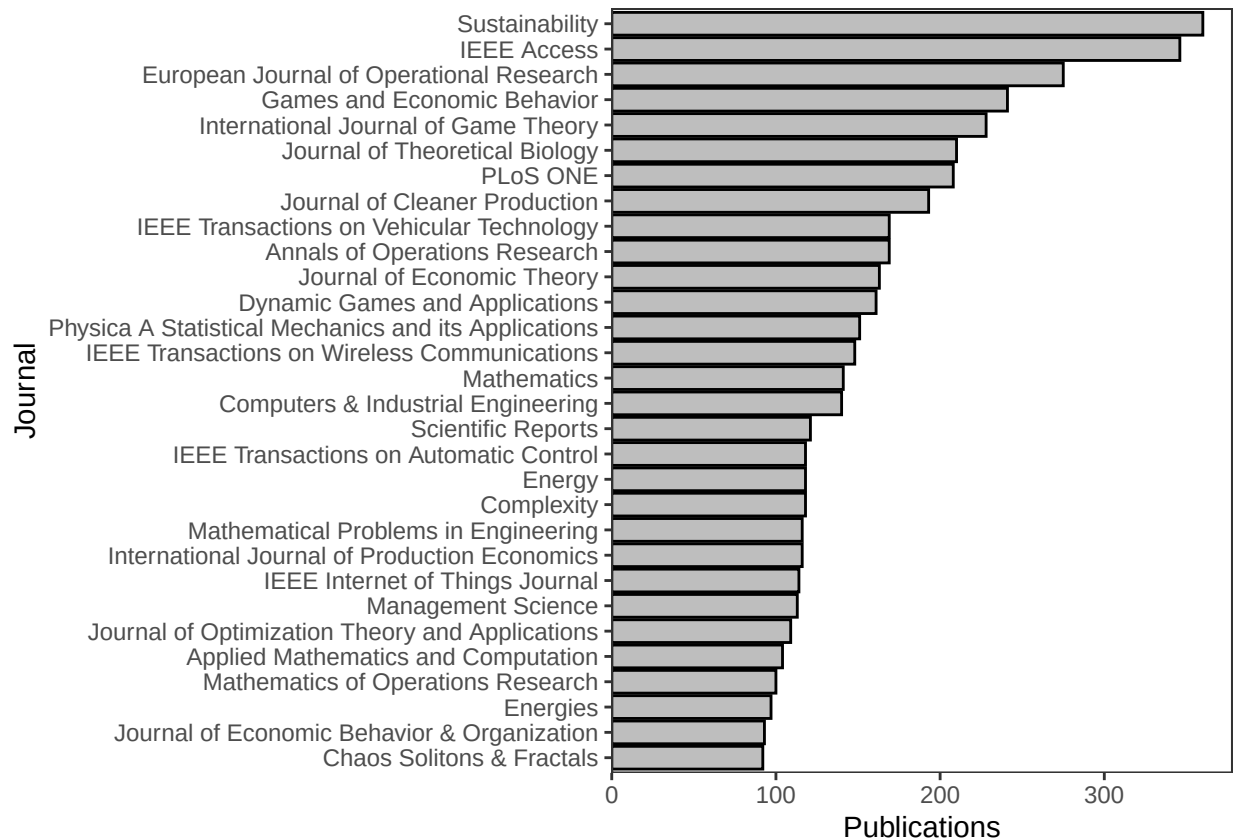
Table 19: Top Authors in Terms of Citations per Discipline

Discipline	Authors		
Arts	Chunli Fu (17)	Keyong Zhang (17)	Ruiyuan Cong (17)
Biology	Mart R. Gross (1681)	John M. McNamara (1531)	Alasdair I. Houston (1425)
Biomedical Research	Martin A. Nowak (11850)	Hisashi Ohtsuki (4001)	Matjaž Perc (3748)
Chemistry	Han Yue Zhang (363)	Ping Ping Shi (363)	Ren Gen Xiong (363)
Computer Science	Xu Chen (1692)	Tansu Alpcan (1299)	Dusit Niyato (1272)
Economics	David M. Kreps (5769)	Robert Wilson (5687)	Eric Maskin (5025)
Engineering and Technology	Dusit Niyato (4390)	Zhu Han (3779)	Tamer Başar (3316)
Geoscience	Lang Xu (687)	Kaveh Madani (569)	Daming You (484)
Health	Amori Yee Mikami (84)	Keith Humphreys (84)	Margaret A. Boucher (84)
Humanities	Robert van Rooy (272)	Bryce Huebner (225)	Mark G. Chen (164)
Law	John T. Scholz (270)	Jack Knight (158)	Lee Epstein (158)
Management	Arvind Parkhe (3117)	Rajiv Lal (1604)	Jan W. Rivkin (1292)
Mathematics	Xiaotie Deng (899)	Alvin E. Roth (895)	Francisco C. Santos (799)
Medicine	Daeyeol Lee (1129)	Dominic J. Barraclough (825)	Michelle Conroy (696)
Operations Research	Michael G.H. Bell (686)	Jun Zhuang (625)	Vicki M. Bier (429)
Other Professional Fields	Anna Nagurney (241)	Yong He (226)	Huseyin Cavusoglu (225)
Other Social Sciences	Eddie C.M. Hui (426)	Jun Zhuang (328)	Adam Millard Ball (291)
Physics	Matjaž Perc (3075)	Zhen Wang (2326)	Attila Szolnoki (2223)
Political Science	Barry R. Weingast (1776)	John H. Aldrich (1210)	David Austen-Smith (980)
Psychology	Peter Hammerstein (1263)	Linda Mealey (984)	Ronald No (882)
Sociology	Gerry Mackie (765)	Elisa Jayne Bienenstock (375)	Phillip Bonacich (375)

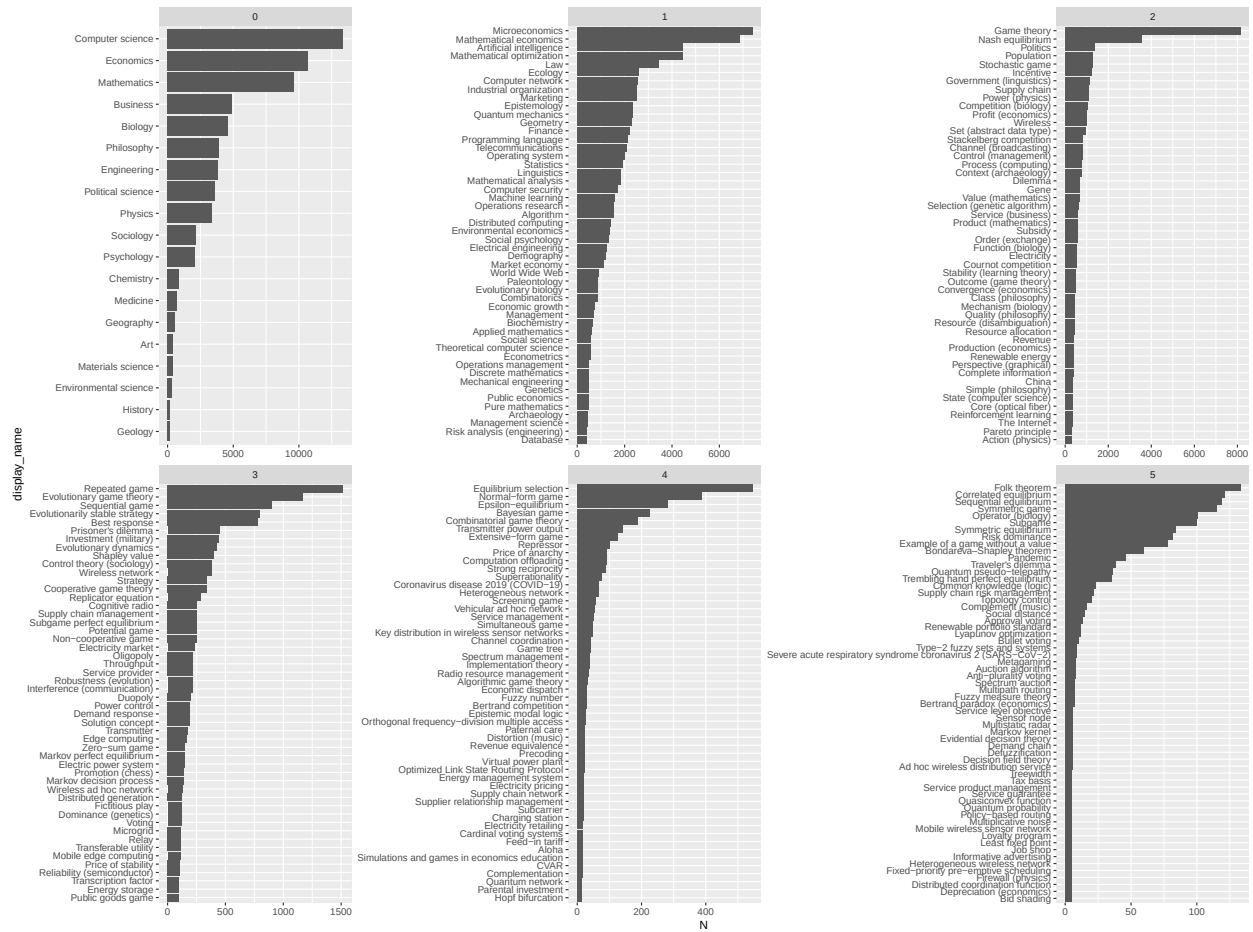
Table 20: Top Authors in Terms of Publications per Subfield

Discipline	Authors		
Arts	Amin Mahmoudi (1)	Aoxuan Jin (1)	Chaoyong Zhang (1)
Biology	Joel S. Brown (10)	Gordon G. McNickle (6)	Mark Broom (6)
Biomedical Research	Martin A. Nowak (53)	Arne Traulsen (24)	Ross Cressman (18)
Chemistry	André Jalles Monteiro (2)	Antônio José Melo Leite (2)	José Nunes da Silva (2)
Computer Science	Kenli Li (14)	Keqin Li (14)	Qiang Zhang (14)
Economics	Drew Fudenberg (20)	Ehud Lehrer (18)	René van den Brink (14)
Engineering and Technology	Dusit Niyato (41)	Zhu Han (41)	H. Vincent Poor (30)
Geoscience	Reza Kerachian (11)	Hong Chen (7)	Zhisong Chen (7)
Health	Lilong Zhu (3)	Hualei Yang (2)	June Liu (2)
Humanities	Michael Franke (4)	Ahti Veikko Pietarinen (3)	Anton Benz (2)
Law	George Halkos (2)	Johannes Urpelainen (2)	Qianqian Li (2)
Management	Ruiliang Yan (13)	Jun Zhuang (11)	S.H. Tijs (11)
Mathematics	Andrzej S. Nowak (20)	Junhai Ma (15)	O. J. Vrieze (15)
Medicine	Amnon Sonnenberg (4)	Daeyeol Lee (3)	A Pandya (2)
Operations Research	Gerhard Wilhelm Weber (9)	Jun Zhuang (8)	Sırma Zeynep Alparslan Gök (8)
Other Professional Fields	Hai Yang (4)	Xiqun Chen (4)	Andrew B. Whinston (3)
Other Social Sciences	Jun Zhuang (7)	Mohammad Taleai (4)	Somaie Abolhasani (4)
Physics	Zhen Wang (21)	Matjaž Perc (20)	Jun Tanimoto (13)
Political Science	Johannes Urpelainen (11)	Christophe Crombez (6)	Ahmer Tarar (5)
Psychology	Andrew M. Colman (6)	Joachim I. Krueger (6)	Lee Alan Dugatkin (5)
Sociology	Elisa Jayne Bienenstock (4)	Phillip Bonacich (4)	Werner Raub (4)

8. Journal Overview



9. Distribution of Concepts



10. Keywords

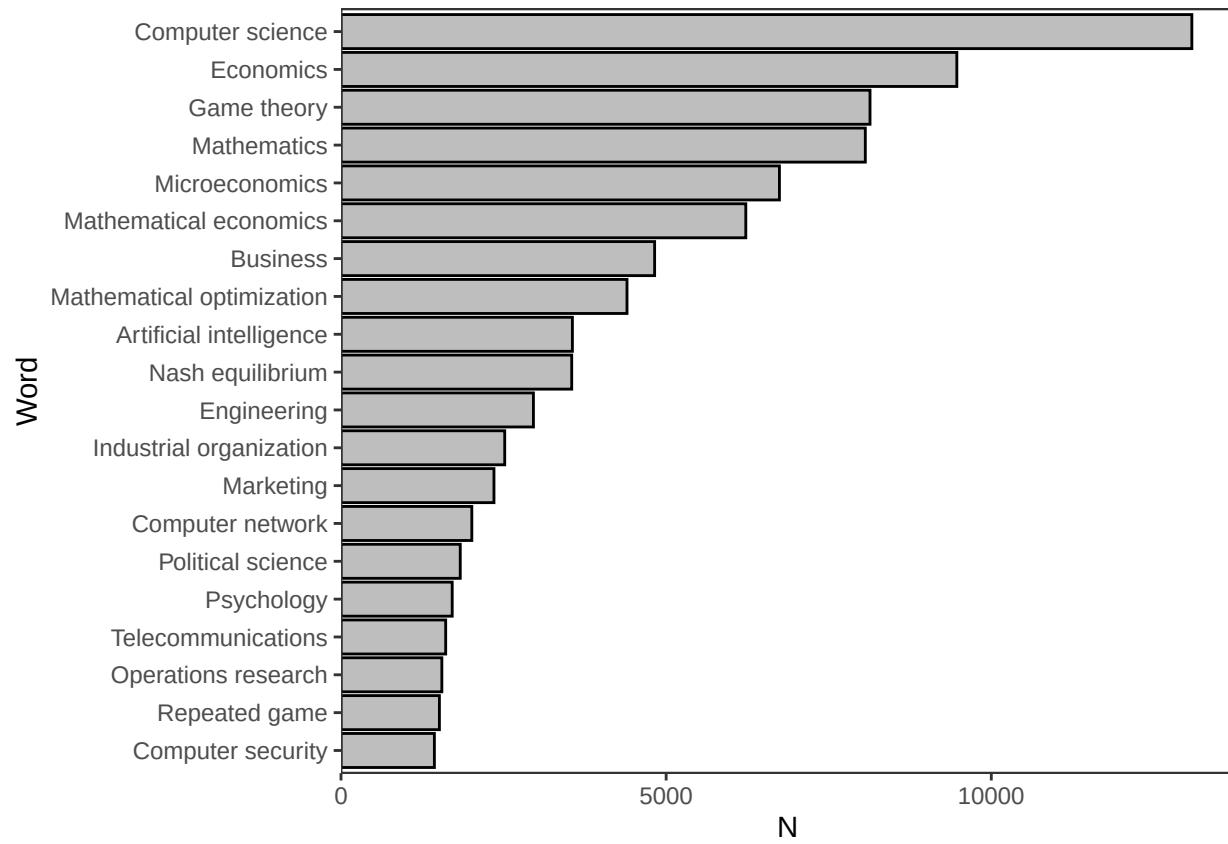


Table 21: Top Keywords

keywords	N
Computer science	13084
Economics	9470
Game theory	8135
Mathematics	8063
Microeconomics	6742
Mathematical economics	6225
Business	4823
Mathematical optimization	4398
Artificial intelligence	3558
Nash equilibrium	3548
Engineering	2959
Industrial organization	2518
Marketing	2352
Computer network	2012
Political science	1834
Psychology	1710
Telecommunications	1610
Operations research	1551
Repeated game	1513
Computer security	1436